

Questions

19M.1.SL.TZ2.18

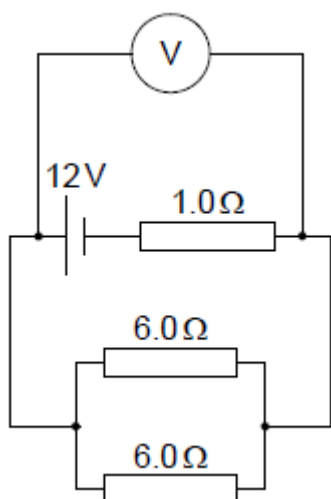
A particle with a charge ne is accelerated through a potential difference V .

What is the magnitude of the work done on the particle?

- A. eV
- B. neV
- C. $\frac{nV}{e}$
- D. $\frac{eV}{n}$

19M.1.SL.TZ2.20

Three resistors of resistance $1.0\ \Omega$, $6.0\ \Omega$ and $6.0\ \Omega$ are connected as shown. The voltmeter is ideal and the cell has an emf of $12\ \text{V}$ with negligible internal resistance.



What is the reading on the voltmeter?

- A. $3.0\ \text{V}$
- B. $4.0\ \text{V}$
- C. $8.0\ \text{V}$
- D. $9.0\ \text{V}$

19M.1.HL.TZ2.28

A transformer with 600 turns in the primary coil is used to change an alternating root mean square (rms) potential difference of $240\ \text{V}_{\text{rms}}$ to $12\ \text{V}_{\text{rms}}$.

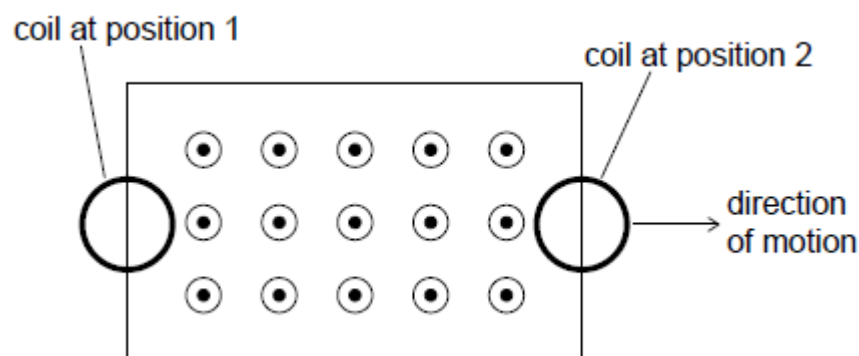
When connected to the secondary coil, a lamp labelled “120 W, 12 V” lights normally. The current in the primary coil is 0.60 A when the lamp is lit.

What are the number of secondary turns and the efficiency of the transformer?

	Number of secondary turns	Efficiency
A.	12000	99 %
B.	30	99 %
C.	12000	83 %
D.	30	83 %

19M.1.HL.TZ2.29

A circular coil of wire moves through a region of uniform magnetic field directed out of the page.



What is the direction of the induced conventional current in the coil for the marked positions?

	Position 1	Position 2
A.	clockwise	clockwise
B.	counterclockwise	clockwise
C.	clockwise	counterclockwise
D.	counterclockwise	counterclockwise

19M.1.HL.TZ2.30

An electron is fixed in position in a uniform electric field. What is the position for which the electrical potential energy of the electron is greatest?



19M.1.HL.TZ2.31

A proton of velocity v enters a region of electric and magnetic fields. The proton is not deflected. An electron and an alpha particle enter the same region with velocity v . Which is correct about the paths of the electron and the alpha particle?

	Path of electron	Path of alpha particle
A.	deflected	deflected
B.	deflected	not deflected
C.	not deflected	deflected
D.	not deflected	not deflected

19M.1.SL.TZ2.22

A particle of mass 0.02 kg moves in a horizontal circle of diameter 1 m with an angular velocity of $3\pi \text{ rad s}^{-1}$.

What is the magnitude and direction of the force responsible for this motion?

	Magnitude of force / N	Direction of force
A.	0.03π	away from centre of circle
B.	0.03π	towards centre of circle
C.	$0.09\pi^2$	away from centre of circle
D.	$0.09\pi^2$	towards centre of circle

19M.1.SL.TZ2.24

A radioactive nuclide with atomic number Z undergoes a process of beta-plus (β^+) decay. What is the atomic number for the nuclide produced and what is another particle emitted during the decay?

	Atomic number	Particle
A.	$Z - 1$	neutrino
B.	$Z + 1$	neutrino
C.	$Z - 1$	anti-neutrino
D.	$Z + 1$	anti-neutrino

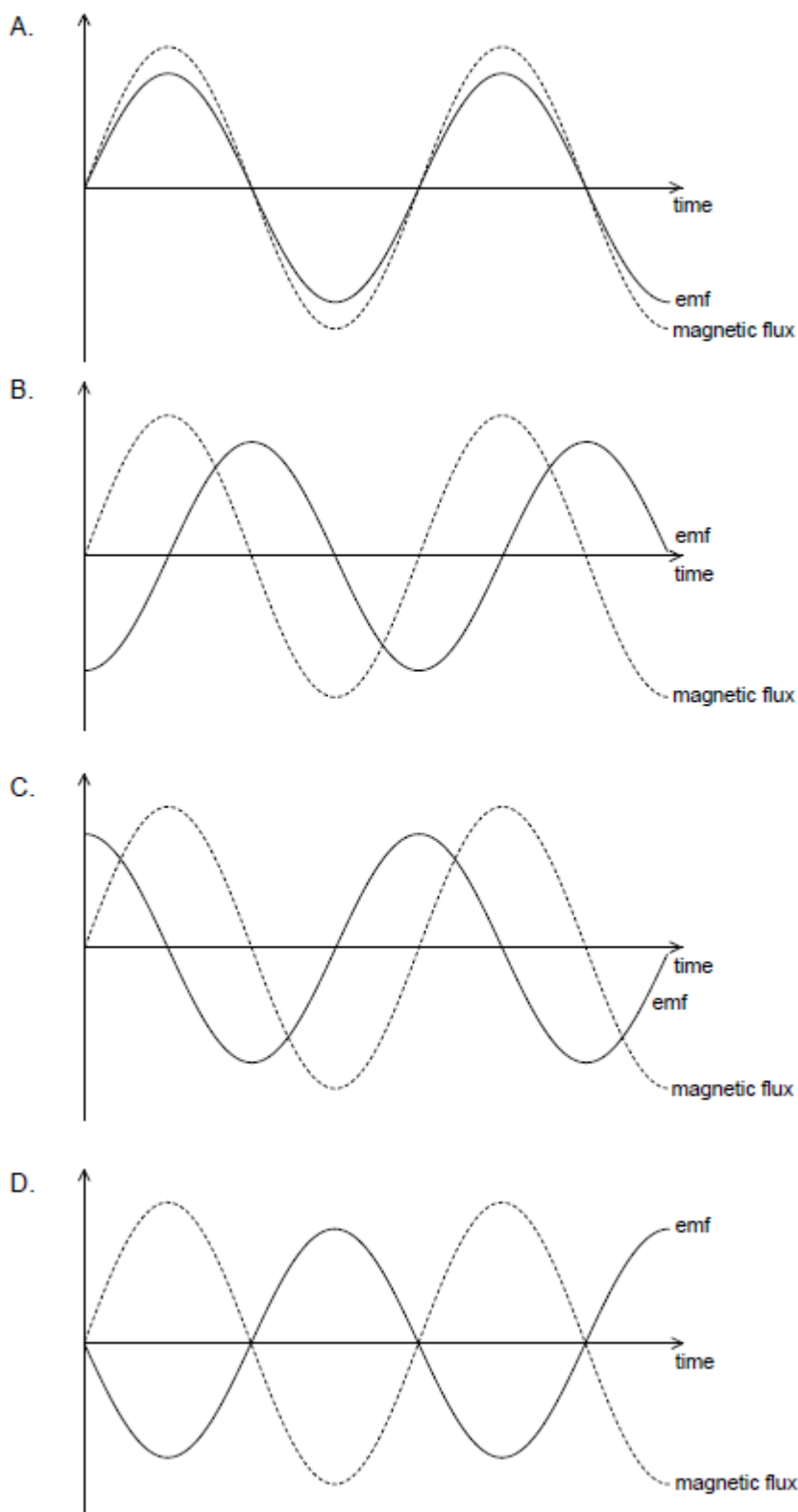
19M.1.SL.TZ2.30

The orbital radius of the Earth around the Sun is 1.5 times that of Venus. What is the intensity of solar radiation at the orbital radius of Venus?

- A. 0.6 kW m^{-2}
- B. 0.9 kW m^{-2}
- C. 2 kW m^{-2}
- D. 3 kW m^{-2}

19N.1.HL.TZ0.34

A coil is rotated in a uniform magnetic field. An alternating emf is induced in the coil. What is a possible phase relationship between the magnetic flux through the coil and the induced emf in the coil when the variations of both quantities are plotted with time?



19N.1.HL.TZ0.40

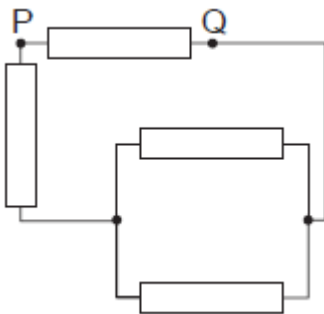
A pure sample of a radioactive nuclide contains N_0 atoms at time $t = 0$. At time t , there are N atoms of the nuclide remaining in the sample. The half-life of the nuclide is $t_{\frac{1}{2}}$.

What is the decay rate of this sample proportional to?

- A. N
- B. $N_0 - N$
- C. t
- D. t_1

20N.1.SL.TZ0.21

Four resistors of $4\ \Omega$ each are connected as shown.

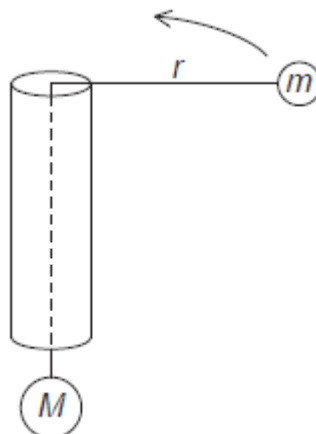


What is the effective resistance between P and Q?

- A. $1.0\ \Omega$
- B. $2.4\ \Omega$
- C. $3.4\ \Omega$
- D. $4.0\ \Omega$

20N.1.SL.TZ0.22

Mass M is attached to one end of a string. The string is passed through a hollow tube and mass m is attached to the other end. Friction between the tube and string is negligible.



Mass m travels at constant speed v in a horizontal circle of radius r . What is mass M ?

- A. $\frac{mv^2}{r}$
- B. mv^2rg
- C. $\frac{mgv^2}{r}$
- D. $\frac{mv^2}{gr}$

20N.1.SL.TZ0.23

Planet X has a gravitational field strength of 18 N kg^{-1} at its surface. Planet Y has the same density as X but three times the radius of X. What is the gravitational field strength at the surface of Y?

- A. 6 m s^{-2}
- B. 18 m s^{-2}
- C. 54 m s^{-2}
- D. 162 m s^{-2}

20N.1.SL.TZ0.18

A metal wire has n free charge carriers per unit volume. The charge on the carrier is q . What additional quantity is needed to determine the current per unit area in the wire?

- A. Cross-sectional area of the wire
- B. Drift speed of charge carriers
- C. Potential difference across the wire
- D. Resistivity of the metal

20N.1.SL.TZ0.24

What are the principal roles of a moderator and of a control rod in a thermal nuclear reactor?

	Role of moderator	Role of control rod
A.	increases kinetic energy of neutrons	maintains a constant rate of reaction
B.	increases kinetic energy of neutrons	absorbs energy transferred in the reactor
C.	reduces kinetic energy of neutrons	maintains a constant rate of reaction
D.	reduces kinetic energy of neutrons	absorbs energy transferred in the reactor

20N.1.SL.TZ0.25

A nuclear power station contains an alternating current generator. What energy transfer is performed by the generator?

- A. Electrical to kinetic
- B. Kinetic to electrical
- C. Nuclear to kinetic
- D. Nuclear to electrical

20N.1.SL.TZ0.19

An electric motor raises an object of weight 500 N through a vertical distance of 3.0 m in 1.5 s. The current in the electric motor is 10 A at a potential difference of 200 V. What is the efficiency of the electric motor?

- A. 17 %
- B. 38 %
- C. 50 %
- D. 75 %

20N.1.SL.TZ0.26

The average temperature of the surface of a planet is five times greater than the average temperature of the surface of its moon. The emissivities of the planet and the moon are the same. The average intensity radiated by the planet is I . What is the average intensity radiated by its moon?

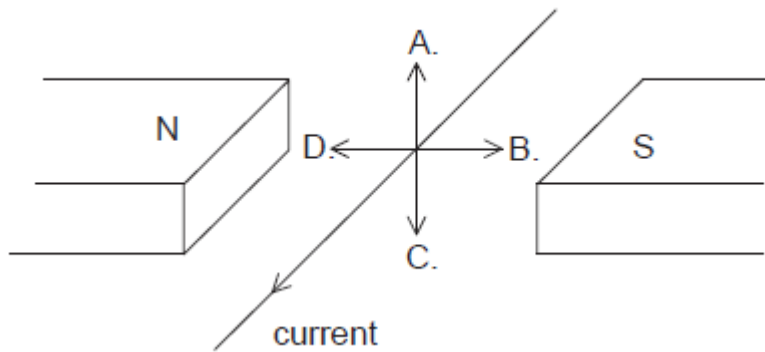
- A. $\frac{I}{25}$
- B. $\frac{I}{125}$

C. $\frac{I}{625}$

D. $\frac{I}{3125}$

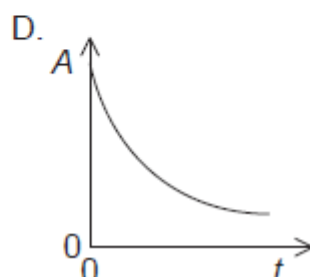
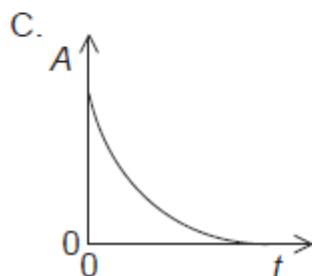
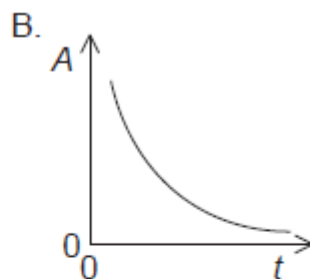
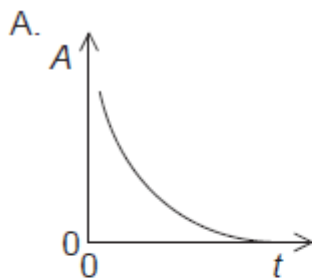
20N.1.SL.TZ0.20

A current in a wire lies between the poles of a magnet. What is the direction of the electromagnetic force on the wire?



20N.1.SL.TZ0.27

Which graph shows the variation of activity A with time t for a radioactive nuclide?



20N.1.SL.TZ0.28

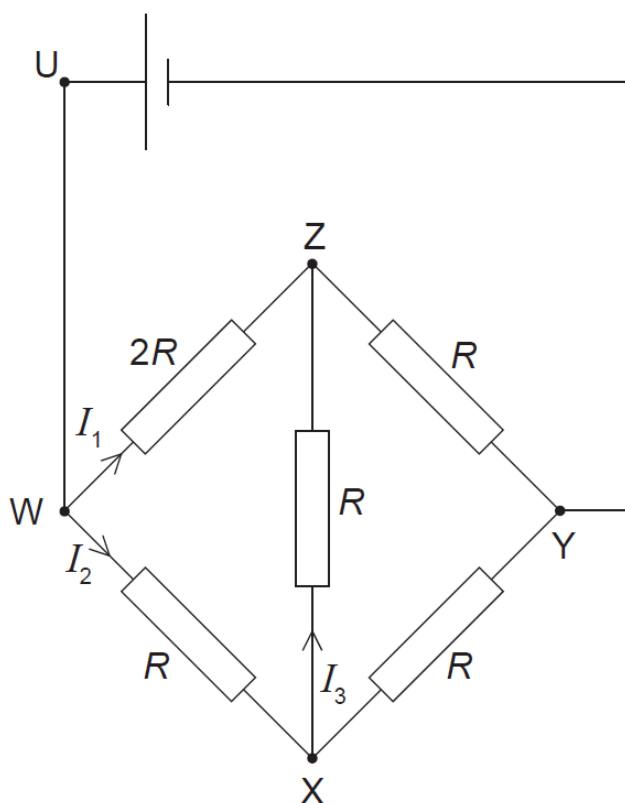
What statement about alpha particles, beta particles and gamma radiation is true?

A. Gamma radiation always travels faster than beta particles in a vacuum.

- B. In air, beta particles produce more ions per unit length travelled than alpha particles.
- C. Alpha particles are always emitted when beta particles are emitted.
- D. Alpha particles are deflected in the same direction as beta particles in a magnetic field.

20N.1.HL.TZ0.15

A cell of electromotive force (emf) E and zero internal resistance is in the circuit shown.



What is correct for loop WXYUW?

- A. $E = 3I_1R - I_3R$
- B. $E = I_3R - I_2R$
- C. $E = I_3R$
- D. $E = 2I_2R - I_3R$

20N.1.HL.TZ0.16

What is the relationship between the resistivity ρ of a uniform wire, the radius r of the wire and the length l of the wire when its resistance is constant?

- A. $\rho \propto r^2l$

B. $\rho \propto r l^2$

C. $\rho \propto \frac{l}{r^2}$

D. $\rho \propto \frac{r^2}{l}$

20N.1.HL.TZ0.17

A power station generates 250 kW of power at a potential difference of 25 kV. The energy is transmitted through cables of total resistance 4.0 Ω .

What is the power loss in the cables?

A. 0.04 kW

B. 0.4 kW

C. 4 kW

D. 40 kW

20N.1.HL.TZ0.24

The mass of nuclear fuel in a nuclear reactor decreases at the rate of 8 mg every hour. The overall reaction process has an efficiency of 50 %. What is the maximum power output of the reactor?

A. 100 MW

B. 200 MW

C. 100 GW

D. 200 GW

20N.1.HL.TZ0.34

Power P is dissipated in a resistor of resistance R when there is a direct current I in the resistor.

What is the average power dissipation in a resistance $\frac{R}{2}$ when the alternating root-mean-square (rms) current in the resistor is $2I$?

- A. P
- B. $2P$
- C. $4P$
- D. $8P$

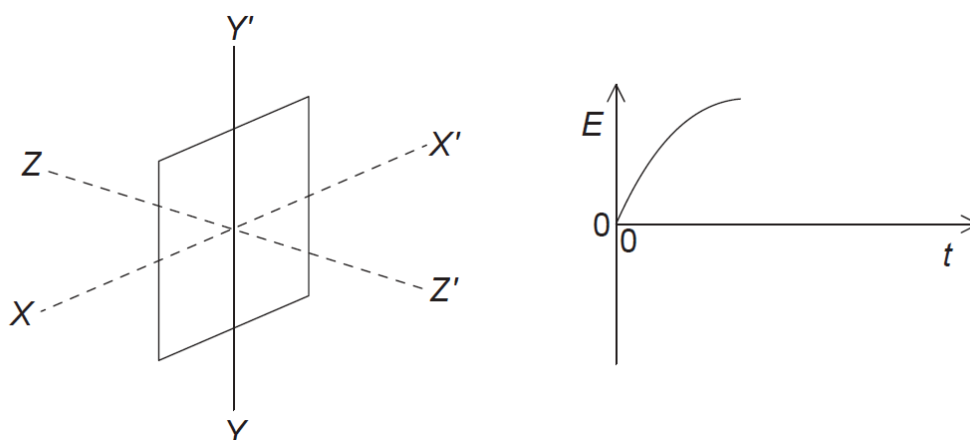
20N.1.HL.TZ0.30

Two satellites W and X have the same mass. They have circular orbits around the same planet. W is closer to the surface than X. What quantity is smaller for W than for X?

- A. Gravitational force from the planet
- B. Angular velocity
- C. Orbital speed
- D. Orbital period

20N.1.HL.TZ0.35

A rectangular coil rotates at a constant angular velocity. At the instant shown, the plane of the coil is at right angles to the line ZZ' . A uniform magnetic field acts in the direction ZZ' .



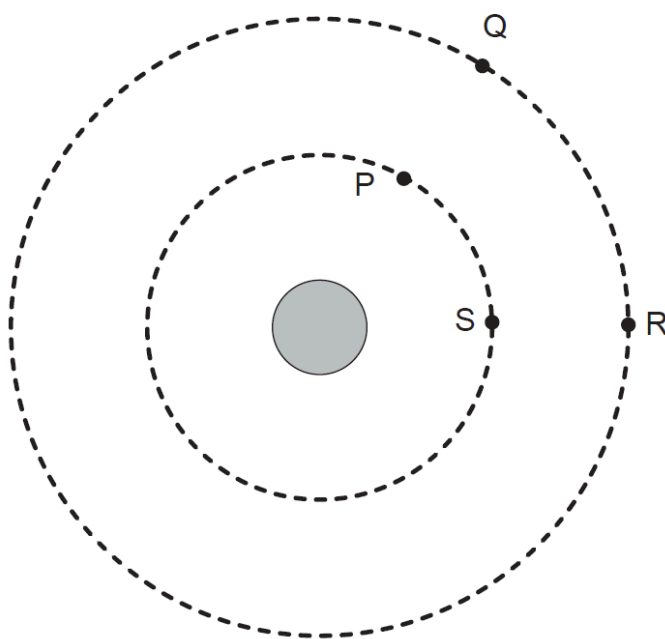
What rotation of the coil about a specified axis will produce the graph of electromotive force (emf) E against time t ?

- A. Through $\frac{\pi}{2}$ about ZZ'

- B. Through π about YY'
- C. Through $\frac{\pi}{2}$ about XX'
- D. Through π about XX'

20N.1.HL.TZ0.31

P and S are two points on a gravitational equipotential surface around a planet. Q and R are two points on a different gravitational equipotential surface at a greater distance from the planet.

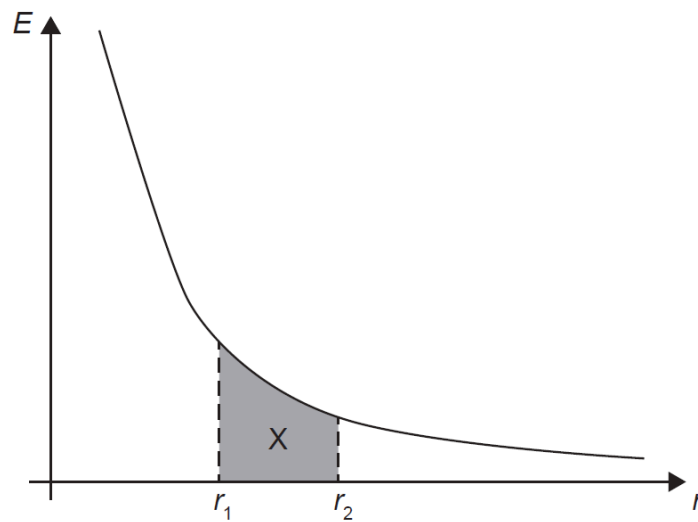


The greatest work done by the gravitational force is when moving a mass from

- A. P to S.
- B. Q to R.
- C. R to P.
- D. S to R.

20N.1.HL.TZ0.32

The graph shows the variation of electric field strength E with distance r from a point charge.



The shaded area X is the area under the graph between two separations r_1 and r_2 from the charge.

What is X?

- A. The electric field average between r_1 and r_2
- B. The electric potential difference between r_1 and r_2
- C. The work done in moving a charge from r_1 to r_2
- D. The work done in moving a charge from r_2 to r_1

20N.1.HL.TZ0.33

Why are high voltages and low currents used when electricity is transmitted over long distances?

- A. Cables can be closer to the ground.
- B. Electrons have a greater drift speed.
- C. Energy losses are reduced.
- D. Resistance of the power lines is reduced.

19N.1.SL.TZ0.22

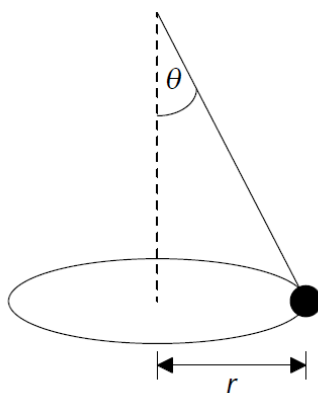
An object of mass m makes n revolutions per second around a circle of radius r at a constant speed. What is the kinetic energy of the object?

- A. 0
- B. $\frac{1}{2}\pi^2 mn^2 r^2$
- C. $2\pi^2 mn^2 r^2$

D. $4\pi^2 mn^2 r^2$

19N.1.SL.TZ0.21

An object hangs from a light string and moves in a horizontal circle of radius r .



The string makes an angle θ with the vertical. The angular speed of the object is ω . What is $\tan \theta$?

A. $\frac{\omega^2 r}{g}$

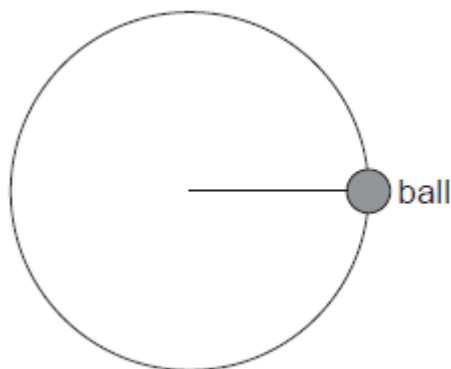
B. $\frac{g}{\omega^2 r}$

C. $\frac{\omega r^2}{g}$

D. $\frac{g}{\omega r^2}$

22M.1.SL.TZ1.23

A ball of mass 0.3 kg is attached to a light, inextensible string. It is rotated in a vertical circle. The length of the string is 0.6 m and the speed of rotation of the ball is 4 m s^{-1} .



What is the tension when the string is horizontal?

A. 5 N

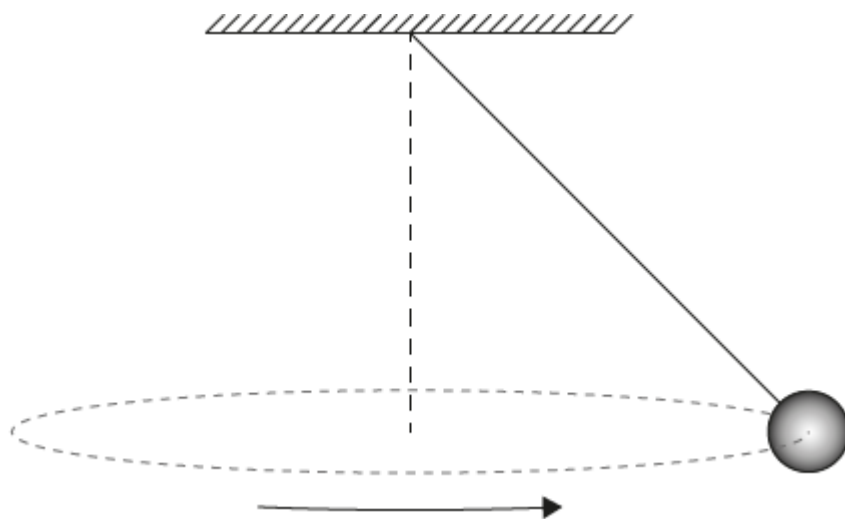
B. 8 N

C. 11 N

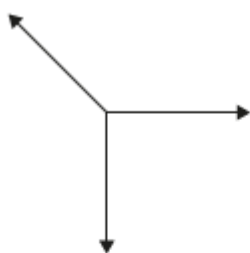
D. 13 N

21M.1.SL.TZ2.24

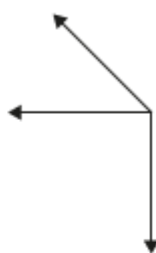
A sphere is suspended from the end of a string and rotates in a horizontal circle. Which freebody diagram, to the correct scale, shows the forces acting on the sphere?



A.



B.



C.



D.



21M.1.SL.TZ2.23

An object moves in a circle of constant radius. Values of the centripetal force F are measured for different values of angular velocity ω . A graph is plotted with ω on the x -axis. Which quantity plotted on the y -axis will produce a straight-line graph?

A. $\sqrt{F}$

- B. F
- C. F^2
- D. $\frac{1}{F}$

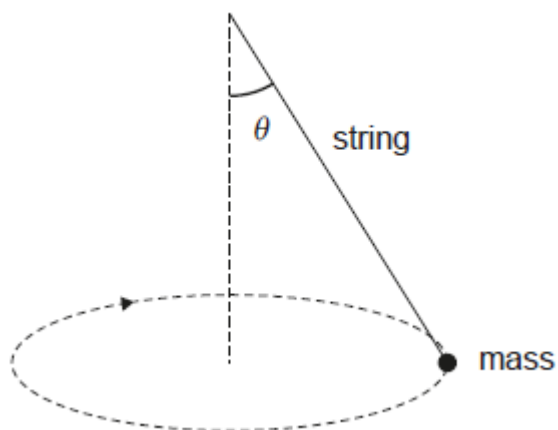
21M.1.SL.TZ2.19

An ion moves in a circle in a uniform magnetic field. Which single change would increase the radius of the circular path?

- A. Decreasing the speed of the ion
- B. Increasing the charge of the ion
- C. Increasing the mass of the ion
- D. Increasing the strength of the magnetic field

21N.1.SL.TZ0.22

A mass at the end of a string is moving in a horizontal circle at constant speed. The string makes an angle θ to the vertical.



What is the magnitude of the acceleration of the mass?

- A. g
- B. $g \sin \theta$
- C. $g \cos \theta$
- D. $g \tan \theta$

22M.1.SL.TZ2.23

A satellite is orbiting Earth in a circular path at constant speed. Three statements about the resultant force on the satellite are:

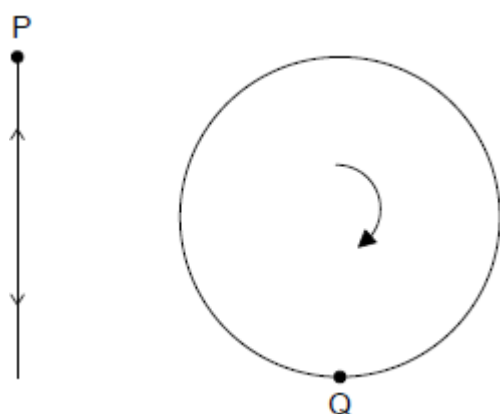
- I. It is equal to the gravitational force of attraction on the satellite.
- II. It is equal to the mass of the satellite multiplied by its acceleration.
- III. It is equal to the centripetal force on the satellite.

Which combination of statements is correct?

- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II and III

19M.1.SL.TZ2.14

Object P moves vertically with simple harmonic motion (shm). Object Q moves in a vertical circle with a uniform speed. P and Q have the same time period T . When P is at the top of its motion, Q is at the bottom of its motion.



What is the interval between successive times when the acceleration of P is equal and opposite to the acceleration of Q?

- A. $\frac{T}{4}$
- B. $\frac{T}{2}$
- C. $\frac{3T}{4}$
- D. T

21M.1.SL.TZ1.22

A child stands on a horizontal rotating platform that is moving at constant angular speed. The centripetal force on the child is provided by

- A. the gravitational force on the child.
- B. the friction on the child's feet.
- C. the tension in the child's muscles.
- D. the normal reaction of the platform on the child.

19M.1.SL.TZ1.25

Satellite X orbits a planet with orbital radius R . Satellite Y orbits the same planet with orbital radius $2R$. Satellites X and Y have the same mass.

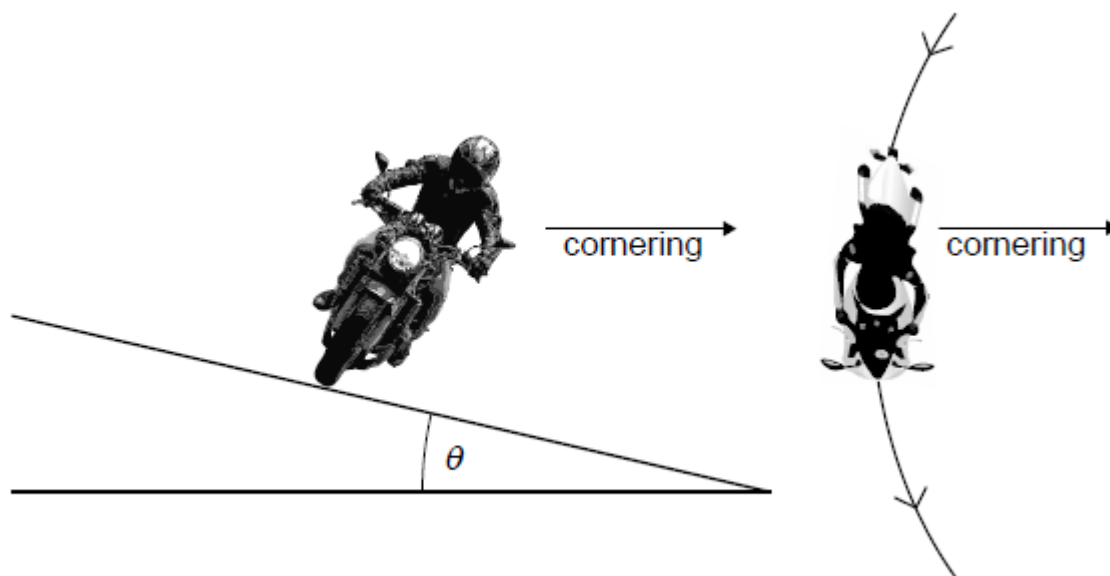
What is the ratio $\frac{\text{centripetal acceleration of X}}{\text{centripetal acceleration of Y}}$?

- A. $\frac{1}{4}$
- B. $\frac{1}{2}$
- C. 2
- D. 4

19M.1.SL.TZ1.24

A motorcyclist is cornering on a curved race track.

Which combination of changes of banking angle θ and coefficient of friction μ between the tyres and road allows the motorcyclist to travel around the corner at greater speed?



	Banking angle θ	Coefficient of friction μ
A.	increase	increase
B.	increase	decrease
C.	decrease	increase
D.	decrease	decrease

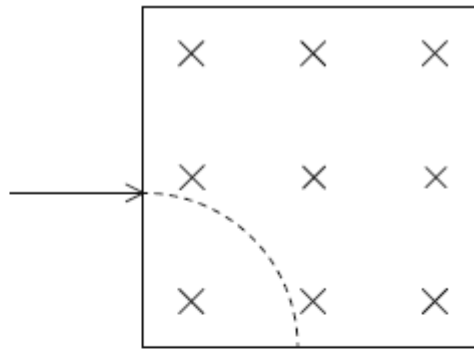
18N.1.HL.TZ0.5

A mass m attached to a string of length R moves in a vertical circle with a constant speed. The tension in the string at the top of the circle is T . What is the kinetic energy of the mass at the top of the circle?

- A. $\frac{R(T + mg)}{2}$
- B. $\frac{R(T - mg)}{2}$
- C. $\frac{Rmg}{2}$
- D. $\frac{R(2T + mg)}{2}$

18N.1.SL.TZ0.22

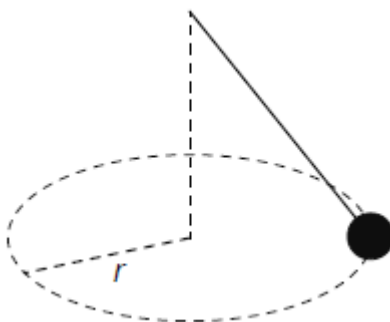
A particle of mass m and charge of magnitude q enters a region of uniform magnetic field B that is directed into the page. The particle follows a circular path of radius R . What are the sign of the charge of the particle and the speed of the particle?



	Charge of the particle	Speed of the particle
A.	positive	$\frac{qBR}{m}$
B.	negative	$\frac{qBR}{m}$
C.	negative	$\sqrt{\frac{qBR}{m}}$
D.	positive	$\sqrt{\frac{qBR}{m}}$

18N.1.SL.TZ0.7

The mass at the end of a pendulum is made to move in a horizontal circle of radius r at constant speed. The magnitude of the net force on the mass is F .



What is the direction of F and the work done by F during half a revolution?

	Direction of F	Work done by F
A.	towards centre of circle	zero
B.	towards centre of circle	$\pi r F$
C.	away from centre of circle	zero
D.	away from centre of circle	$\pi r F$

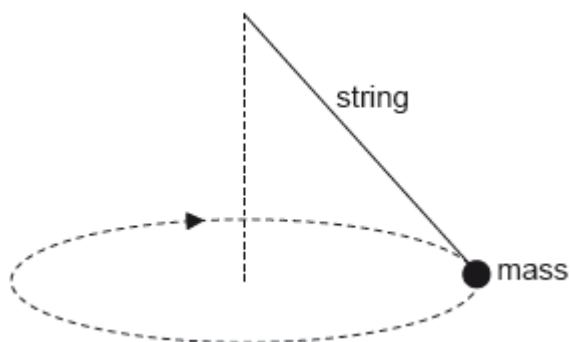
18M.1.HL.TZ2.17

An object of mass m moves in a horizontal circle of radius r with a constant speed v . What is the rate at which work is done by the centripetal force?

- A. $\frac{mv^3}{r}$
- B. $\frac{mv^3}{2\pi r}$
- C. $\frac{mv^3}{4\pi r}$
- D. zero

18M.1.SL.TZ2.23

A mass at the end of a string is swung in a horizontal circle at increasing speed until the string breaks.



The subsequent path taken by the mass is a

- A. line along a radius of the circle.
- B. horizontal circle.
- C. curve in a horizontal plane.
- D. curve in a vertical plane.

17N.1.SL.TZ0.22

A satellite X of mass m orbits the Earth with a period T . What will be the orbital period of satellite Y of mass $2m$ occupying the same orbit as X?

- A. $\frac{T}{2}$
- B. T
- C. $\sqrt{2}T$
- D. $2T$

18M.1.SL.TZ1.22

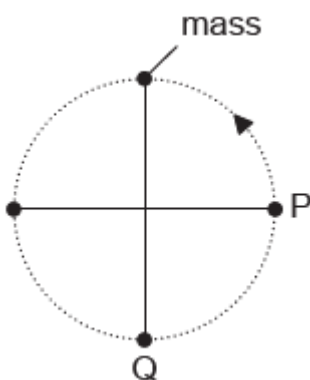
An object of mass m at the end of a string of length r moves in a vertical circle at a constant angular speed ω .

What is the tension in the string when the object is at the bottom of the circle?

- A. $m(\omega^2 r + g)$
- B. $m(\omega^2 r - g)$
- C. $mg(\omega^2 r + 1)$
- D. $mg(\omega^2 r - 1)$

17N.1.SL.TZ0.21

A mass attached to a string rotates in a gravitational field with a constant period in a vertical plane.

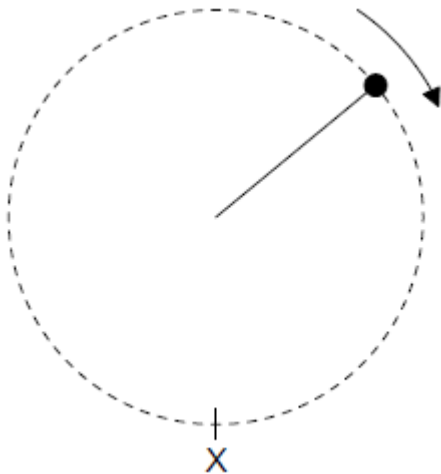


How do the tension in the string and the kinetic energy of the mass compare at P and Q?

	Tension in the string	Kinetic energy of mass
A.	greater at P than Q	greater at Q than P
B.	greater at Q than P	greater at Q than P
C.	greater at P than Q	same at Q and P
D.	greater at Q than P	same at Q and P

17M.1.HL.TZ2.18

A small ball of weight W is attached to a string and moves in a vertical circle of radius R .



What is the smallest kinetic energy of the ball at position X for the ball to maintain the circular motion with radius R ?

- A. $\frac{WR}{2}$
- B. WR
- C. $2WR$
- D. $\frac{5WR}{2}$

17M.1.SL.TZ2.22

Two satellites of mass m and $2m$ orbit a planet at the same orbit radius. If F is the force exerted on the satellite of mass m by the planet and a is the centripetal acceleration of this satellite, what is the force and acceleration of the satellite with mass $2m$?

	Force	Acceleration
A.	$2F$	a
B.	$2F$	$\frac{a}{2}$
C.	F	a
D.	F	$\frac{a}{2}$

16N.1.SL.TZ0.22

An object at the end of a wooden rod rotates in a vertical circle at a constant angular velocity. What is correct about the tension in the rod?

- A. It is greatest when the object is at the bottom of the circle.
- B. It is greatest when the object is halfway up the circle.
- C. It is greatest when the object is at the top of the circle.
- D. It is unchanged throughout the motion.

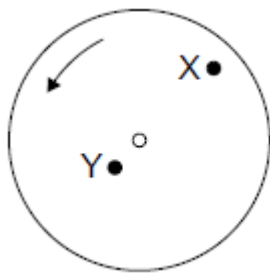
17M.1.SL.TZ1.23

An object of constant mass is tied to the end of a rope of length l and made to move in a horizontal circle. The speed of the object is increased until the rope breaks at speed v . The length of the rope is then changed. At what other combination of rope length and speed will the rope break?

	Rope length	Speed
A.	$4l$	$2v$
B.	$2l$	v
C.	$2l$	$\frac{v}{2}$
D.	$4l$	$\frac{v}{2}$

17M.1.SL.TZ1.22

A horizontal disc rotates uniformly at a constant angular velocity about a central axis normal to the plane of the disc.



Point X is a distance $2L$ from the centre of the disc. Point Y is a distance L from the centre of the disc. Point Y has a linear speed v and a centripetal acceleration a .

What is the linear speed and centripetal acceleration of point X?

	Linear speed of X	Centripetal acceleration of X
A.	v	a
B.	$2v$	$2a$
C.	v	$2a$
D.	$2v$	$4a$

19M.1.SL.TZ2.28

A neutron collides head-on with a stationary atom in the moderator of a nuclear power station. The kinetic energy of the neutron changes as a result. There is also a change in the probability that this neutron can cause nuclear fission.

What are these changes?

	Change in kinetic energy of the neutron	Change in probability of causing nuclear fission
A.	increase	increase
B.	decrease	increase
C.	increase	decrease
D.	decrease	decrease

17M.1.SL.TZ2.28

The main role of a moderator in a nuclear fission reactor is to

- A. slow down neutrons.
- B. absorb neutrons.
- C. reflect neutrons back to the reactor.
- D. accelerate neutrons.

18M.1.HL.TZ2.23

What part of a nuclear power station is principally responsible for increasing the chance that a neutron will cause fission?

- A. Moderator
- B. Control rod
- C. Pressure vessel
- D. Heat exchanger

18M.1.HL.TZ1.23

A nuclear reactor contains atoms that are used for moderation and atoms that are used for control.

What are the ideal properties of the moderator atoms and the control atoms in terms of neutron absorption?

	Ideal moderator atom	Ideal control atom
A.	poor absorber of neutrons	poor absorber of neutrons
B.	poor absorber of neutrons	good absorber of neutrons
C.	good absorber of neutrons	poor absorber of neutrons
D.	good absorber of neutrons	good absorber of neutrons

18N.1.SL.TZ0.28

What is the function of control rods in a nuclear power plant?

- A. To slow neutrons down
- B. To regulate fuel supply

- C. To exchange thermal energy
- D. To regulate the reaction rate

19N.1.SL.TZ0.29

What is the function of the moderator in a thermal nuclear fission reactor?

- A. To decrease the kinetic energy of neutrons emitted from fission reactions
- B. To increase the kinetic energy of neutrons emitted from fission reactions
- C. To decrease the overall number of neutrons available for fission
- D. To increase the overall number of neutrons available for fission

19N.1.HL.TZ0.6

A nuclear particle has an energy of 10^8 eV. A grain of sand has a mass of 32 mg. What speed must the grain of sand have for its kinetic energy to equal the energy of the nuclear particle?

- A. 1 mm s^{-1}
- B. 3 mm s^{-1}
- C. 10 mm s^{-1}
- D. 16 mm s^{-1}

17M.1.SL.TZ1.25

What is the definition of the unified atomic mass unit?

- A. $\frac{1}{12}$ the mass of a neutral atom of carbon-12
- B. The mass of a neutral atom of hydrogen-1
- C. $\frac{1}{12}$ the mass of a nucleus of carbon-12
- D. The mass of a nucleus of hydrogen-1

17M.1.SL.TZ1.26

In nuclear fission, a nucleus of element X absorbs a neutron (n) to give a nucleus of element Y and a nucleus of element Z.



What is $\frac{\text{magnitude of the binding energy per nucleon of Y}}{\text{magnitude of the binding energy per nucleon of X}}$ and $\frac{\text{total binding energy of Y and Z}}{\text{total binding energy of X}}$?

	$\frac{\text{Magnitude of the binding energy per nucleon of Y}}{\text{Magnitude of the binding energy per nucleon of X}}$	$\frac{\text{Total binding energy of Y and Z}}{\text{Total binding energy of X}}$
A.	greater than 1	greater than 1
B.	less than 1	greater than 1
C.	greater than 1	less than 1
D.	less than 1	less than 1

17M.1.SL.TZ1.27

What is the energy equivalent to the mass of one proton?

- A. $9.38 \times (3 \times 10^8)^2 \times 10^6 \text{ J}$
- B. $9.38 \times (3 \times 10^8)^2 \times 1.6 \times 10^{-19} \text{ J}$
- C. $\frac{9.38 \times 10^8}{1.6 \times 10^{-19}} \text{ J}$
- D. $9.38 \times 10^8 \times 1.6 \times 10^{-19} \text{ J}$

21M.1.HL.TZ2.22

During the nuclear fission of nucleus X into nucleus Y and nucleus Z, energy is released. The binding energies per nucleon of X, Y and Z are B_X , B_Y and B_Z respectively. What is true about the binding energy per nucleon of X, Y and Z?

- A. $B_Y > B_X$ and $B_Z > B_X$
- B. $B_X = B_Y$ and $B_X = B_Z$
- C. $B_X > B_Y$ and $B_X > B_Z$
- D. $B_X = B_Y + B_Z$

16N.1.SL.TZ0.25

When an alpha particle collides with a nucleus of nitrogen-14 (${}^{14}_7\text{N}$), a nucleus X can be produced together with a proton. What is X?

- A. ${}^{18}_8\text{X}$
- B. ${}^{17}_8\text{X}$
- C. ${}^{18}_9\text{X}$

D. ${}^{17}_9\text{X}$

16N.1.SL.TZ0.26

The mass defect for deuterium is 4×10^{-30} kg. What is the binding energy of deuterium?

- A. 4×10^{-7} eV
- B. 8×10^{-2} eV
- C. 2×10^6 eV
- D. 2×10^{12} eV

17M.1.SL.TZ2.26

The binding energy per nucleon of ${}^{11}_4\text{Be}$ is 6 MeV. What is the energy required to separate the nucleons of this nucleus?

- A. 24 MeV
- B. 42 MeV
- C. 66 MeV
- D. 90 MeV

17M.1.HL.TZ2.21

In the nuclear reaction $\text{X} + \text{Y} \rightarrow \text{Z} + \text{W}$, involving nuclides X, Y, Z and W, energy is released. Which is correct about the masses (M) and the binding energies (BE) of the nuclides?

	Masses	Binding energies
A.	$M_{\text{X}} + M_{\text{Y}} < M_{\text{Z}} + M_{\text{W}}$	$BE_{\text{X}} + BE_{\text{Y}} < BE_{\text{Z}} + BE_{\text{W}}$
B.	$M_{\text{X}} + M_{\text{Y}} < M_{\text{Z}} + M_{\text{W}}$	$BE_{\text{X}} + BE_{\text{Y}} > BE_{\text{Z}} + BE_{\text{W}}$
C.	$M_{\text{X}} + M_{\text{Y}} > M_{\text{Z}} + M_{\text{W}}$	$BE_{\text{X}} + BE_{\text{Y}} < BE_{\text{Z}} + BE_{\text{W}}$
D.	$M_{\text{X}} + M_{\text{Y}} > M_{\text{Z}} + M_{\text{W}}$	$BE_{\text{X}} + BE_{\text{Y}} > BE_{\text{Z}} + BE_{\text{W}}$

18M.1.SL.TZ1.25

The average binding energy per nucleon of the $^{15}_8\text{O}$ nucleus is 7.5 MeV. What is the total energy required to separate the nucleons of one nucleus of $^{15}_8\text{O}$?

- A. 53 MeV
- B. 60 MeV
- C. 113 MeV
- D. 173 MeV

17N.1.SL.TZ0.24

What gives the total change in nuclear mass and the change in nuclear binding energy as a result of a nuclear fusion reaction?

	Nuclear mass	Nuclear binding energy
A.	decreases	decreases
B.	decreases	increases
C.	increases	decreases
D.	increases	increases

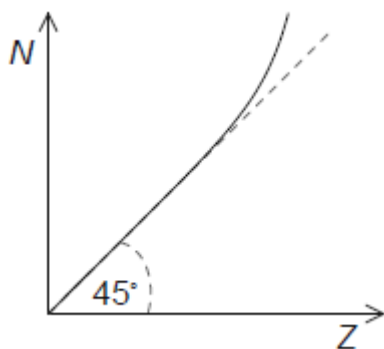
18M.1.SL.TZ2.26

A graph of the variation of average binding energy per nucleon with nucleon number has a maximum. What is indicated by the region around the maximum?

- A. The position below which radioactive decay cannot occur
- B. The region in which fission is most likely to occur
- C. The position where the most stable nuclides are found
- D. The region in which fusion is most likely to occur

18N.1.SL.TZ0.25

The graph shows the variation of the number of neutrons N with the atomic number Z for stable nuclei. The same scale is used in the N and Z axes.



Which information can be inferred from the graph?

- I. For stable nuclei with high Z , N is larger than Z .
- II. For stable nuclei with small Z , $N = Z$.
- III. All stable nuclei have more neutrons than protons.

- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II and III

19M.1.SL.TZ1.26

Which property of a nuclide does **not** change as a result of beta decay?

- A. Nucleon number
- B. Neutron number
- C. Proton number
- D. Charge

19M.1.SL.TZ1.27

The rest mass of the helium isotope ${}^3_2\text{He}$ is m .

Which expression gives the binding energy per nucleon for ${}^3_2\text{He}$?

- A. $\frac{(2m_p + m_n + m)c^2}{3}$
- B. $\frac{(2m_p + m_n - m)c^2}{3}$
- C. $(2m_p + m_n + m)c^2$

D. $(2m_p + m_n - m)c^2$

21M.1.SL.TZ1.25

What is the relation between the value of the unified atomic mass unit in grams and the value of Avogadro's constant in mol^{-1} ?

- A. Their ratio is 1.
- B. Their product is 1.
- C. Their sum is 1.
- D. Their difference is 0.

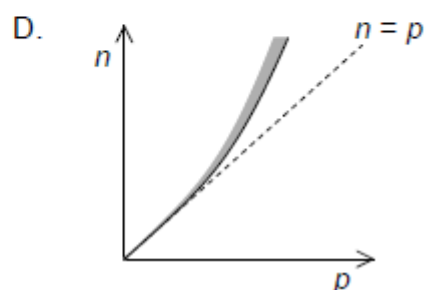
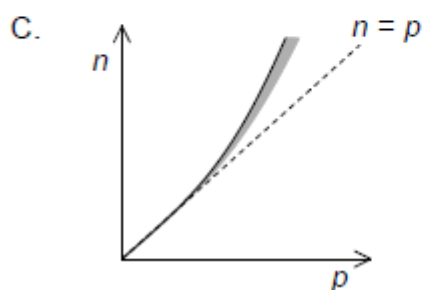
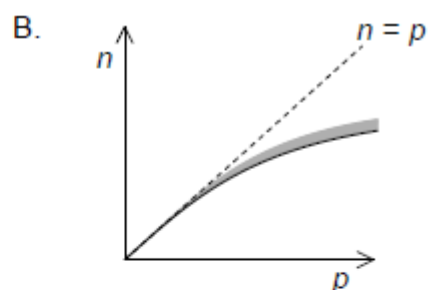
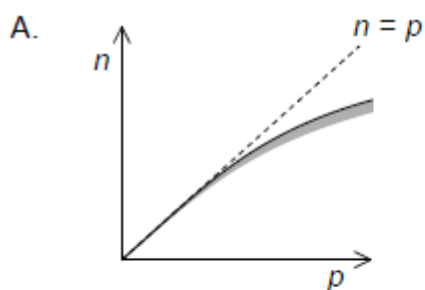
21M.1.HL.TZ1.22

In a hydrogen atom, the sum of the masses of a proton and of an electron is larger than the mass of the atom. Which interaction is mainly responsible for this difference?

- A. Electromagnetic
- B. Strong nuclear
- C. Weak nuclear
- D. Gravitational

19M.1.SL.TZ2.25

The positions of stable nuclei are plotted by neutron number n and proton number p . The graph indicates a dotted line for which $n = p$. Which graph shows the line of stable nuclides and the shaded region where unstable nuclei emit beta minus (β^-) particles?



22M.1.HL.TZ2.24

A neutron is absorbed by a nucleus of uranium-235 $^{235}_{92}\text{U}$. One possible outcome is the production of two nuclides, barium-144 $^{144}_{56}\text{Ba}$ and krypton-89 $^{89}_{36}\text{Kr}$.

How many neutrons are released in this reaction?

- A. 0
- B. 1
- C. 2
- D. 3

21N.1.SL.TZ0.25

The mass of a nucleus of iron-56 ($^{56}_{26}\text{Fe}$) is M .

What is the mass defect of the nucleus of iron-56?

- A. $M - 26m_p - 56m_n$
- B. $26m_p + 30m_n - M$
- C. $M - 26m_p - 56m_n - 26m_e$
- D. $26m_p + 30m_n + 26m_e - M$

22M.1.HL.TZ2.38

Samples of two radioactive nuclides X and Y are held in a container. The number of particles of X is half the number of particles of Y. The half-life of X is twice the half-life of Y.

What is the initial value of $\frac{\text{activity of radioisotope X}}{\text{activity of radioisotope Y}}$?

- A. $\frac{1}{4}$
- B. $\frac{1}{2}$
- C. 1
- D. 4

22M.1.HL.TZ1.40

The decay constant, λ , of a radioactive sample can be defined as

- A. the number of disintegrations in the radioactive sample.
- B. the number of disintegrations per unit time in the radioactive sample.
- C. the probability that a nucleus decays in the radioactive sample.
- D. the probability that a nucleus decays per unit time in the radioactive sample.

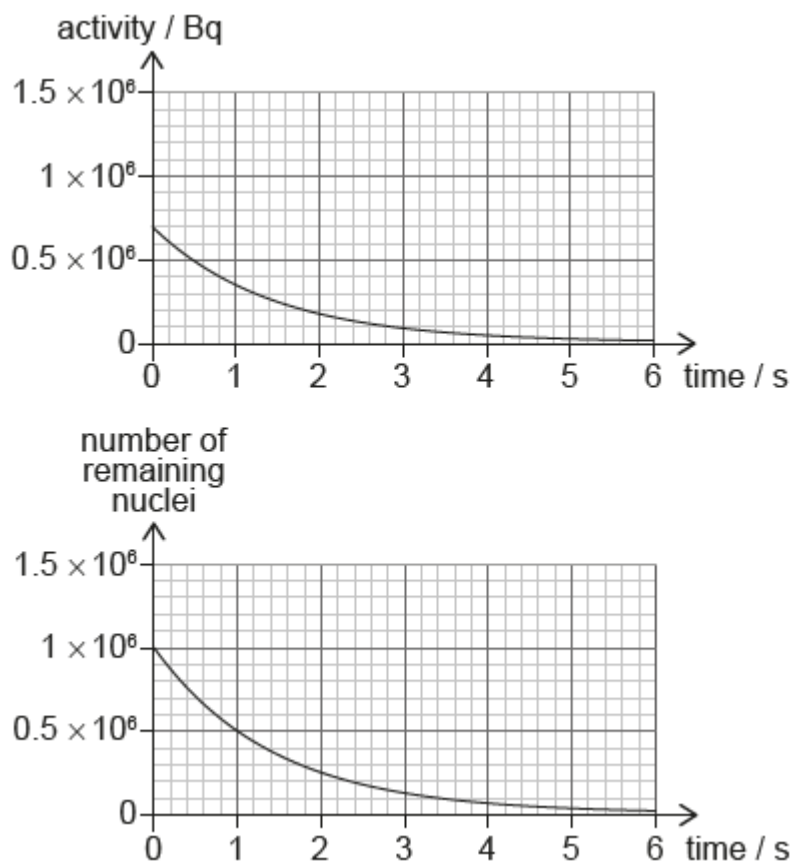
21M.1.HL.TZ1.40

What was a reason to postulate the existence of neutrinos?

- A. Nuclear energy levels had a continuous spectrum.
- B. The photon emission spectrum only contained specific wavelengths.
- C. Some particles were indistinguishable from their antiparticle.
- D. The energy of emitted beta particles had a continuous spectrum.

21M.1.HL.TZ1.39

The graphs show the variation with time of the activity and the number of remaining nuclei for a sample of a radioactive nuclide.



What is the decay constant of the nuclide?

- A. 0.7 s^{-1}
- B. 1 s^{-1}
- C. $\frac{1}{0.7} \text{ s}^{-1}$
- D. 1.5 s^{-1}

19M.1.HL.TZ1.39

The half-life of a radioactive nuclide is 8.0 s. The initial activity of a pure sample of the nuclide is 10 000 Bq. What is the approximate activity of the sample after 4.0 s?

- A. 2500 Bq
- B. 5000 Bq
- C. 7100 Bq
- D. 7500 Bq

18N.1.HL.TZ0.40

A radioactive nuclide is known to have a very long half-life.

Three quantities known for a pure sample of the nuclide are

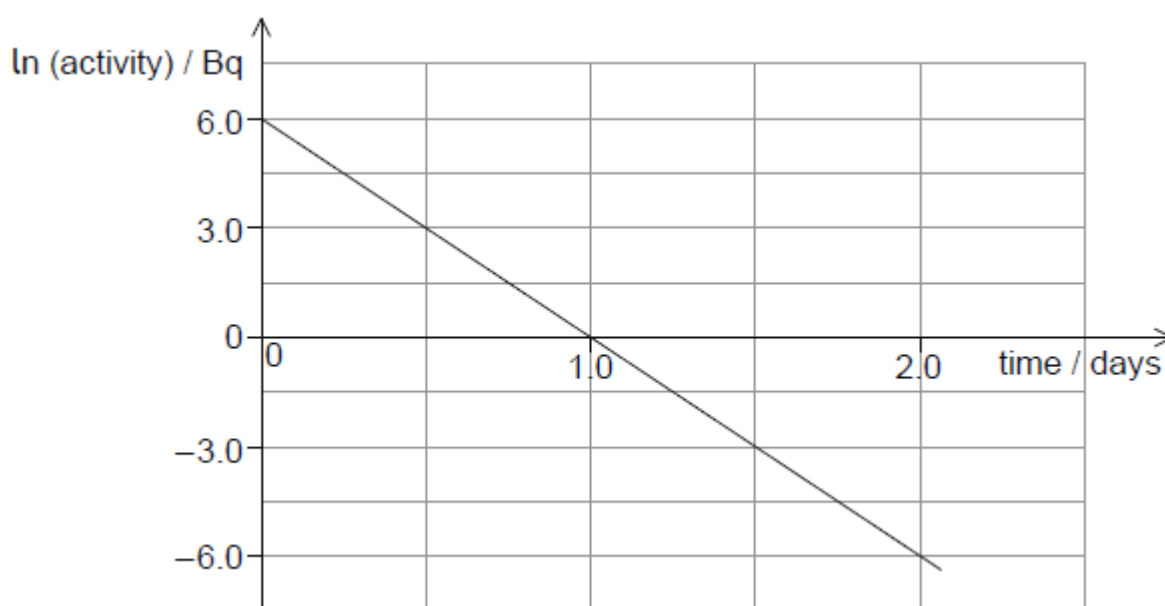
- I. the activity of the nuclide
- II. the number of nuclide atoms
- III. the mass number of the nuclide.

What quantities are required to determine the half-life of the nuclide?

- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II and III

18N.1.HL.TZ0.39

The graph shows the variation of the natural log of activity, $\ln(\text{activity})$, against time for a radioactive nuclide.



What is the decay constant, in days^{-1} , of the radioactive nuclide?

- A. $\frac{1}{6}$
- B. $\frac{1}{3}$
- C. 3
- D. 6

18M.1.HL.TZ2.40

Two samples X and Y of different radioactive isotopes have the same initial activity. Sample X has twice the number of atoms as sample Y. The half-life of X is T . What is the half-life of Y?

- A. $2T$
- B. T
- C. $\frac{T}{2}$
- D. $\frac{T}{4}$

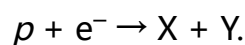
17M.1.HL.TZ2.40

A radioactive element has decay constant λ (expressed in s^{-1}). The number of nuclei of this element at $t = 0$ is N . What is the expected number of nuclei that will have decayed after 1 s?

- A. $N(1 - e^{-\lambda})$
- B. $\frac{N}{\lambda}$
- C. $Ne^{-\lambda}$
- D. λN

17M.1.HL.TZ1.40

Electron capture can be represented by the equation

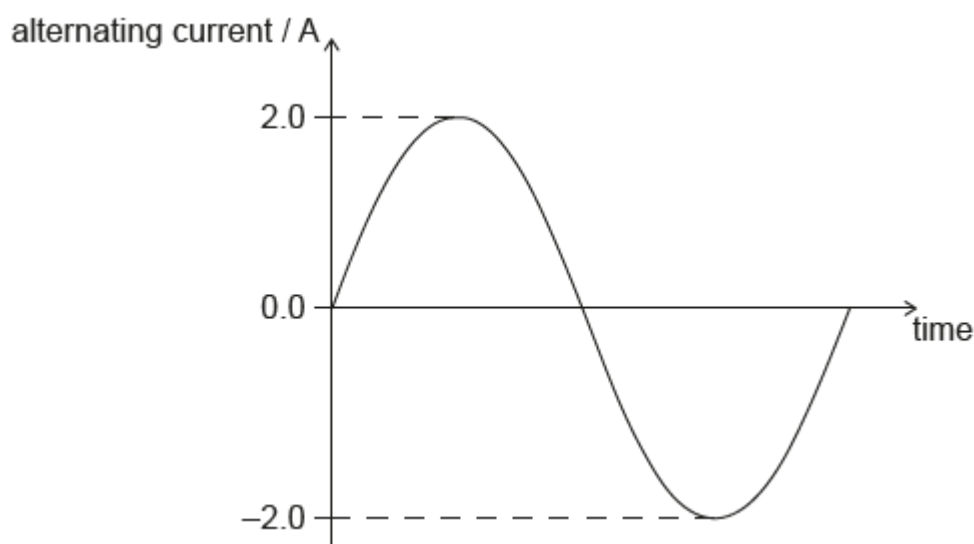


What are X and Y?

	X	Y
A.	proton	positron
B.	electron	positron
C.	neutron	electron antineutrino
D.	neutron	electron neutrino

21M.1.HL.TZ2.34

The graph shows the variation of an alternating current with time in a $4.0\ \Omega$ resistor.



What is the average power dissipated in the resistor?

- A. 4 W
- B. 8 W
- C. 16 W
- D. 32 W

16N.1.HL.TZ0.33

Which of the following reduces the energy losses in a transformer?

- A. Using thinner wires for the windings.
- B. Using a solid core instead of a laminated core.
- C. Using a core made of steel instead of iron.
- D. Linking more flux from the primary to the secondary core.

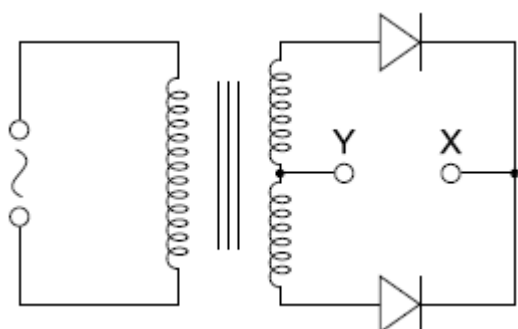
17M.1.HL.TZ1.32

A direct current (dc) of 5A dissipates a power P in a resistor. Which peak value of the alternating current (ac) will dissipate an average power P in the same resistor?

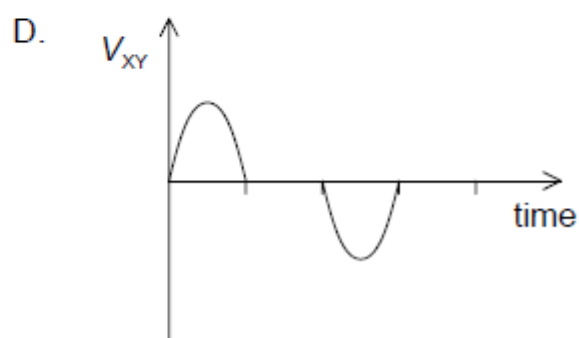
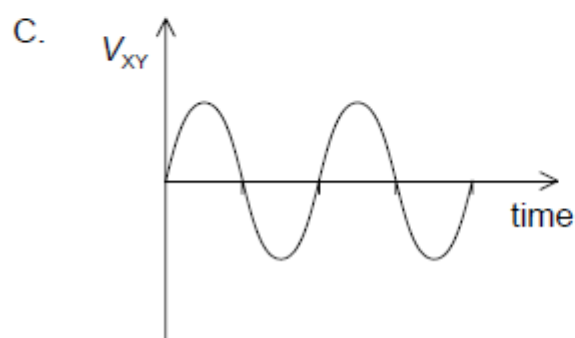
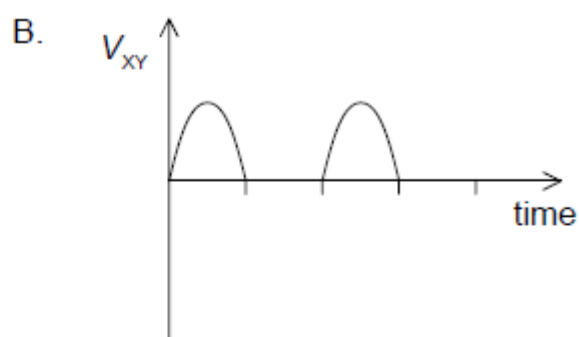
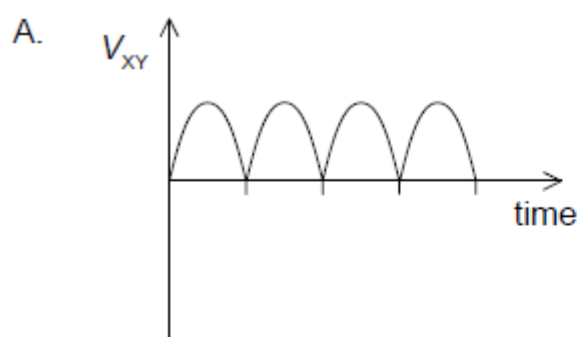
- A. 5A
- B. $\frac{5}{2}$ A
- C. $\frac{5}{\sqrt{2}}$ A
- D. $5\sqrt{2}$ A

16N.1.HL.TZ0.34

The secondary coil of an alternating current (ac) transformer is connected to two diodes as shown.



Which graph shows the variation with time of the potential difference V_{XY} between X and Y?



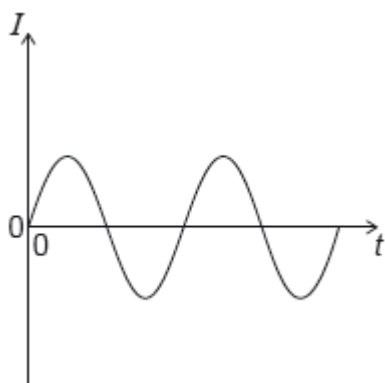
17N.1.HL.TZ0.36

An alternating current (ac) generator produces a peak emf E_0 and periodic time T . What are the peak emf and periodic time when the frequency of rotation is doubled?

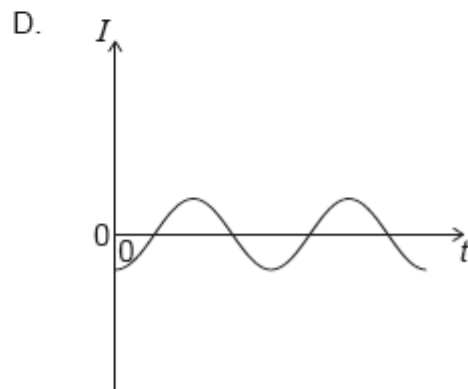
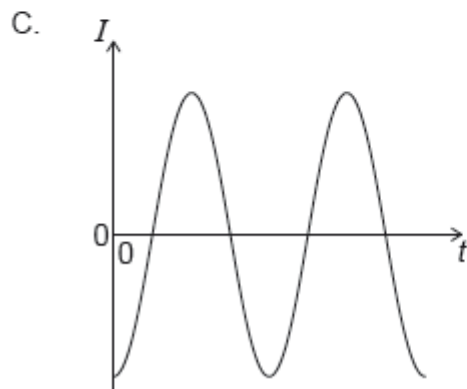
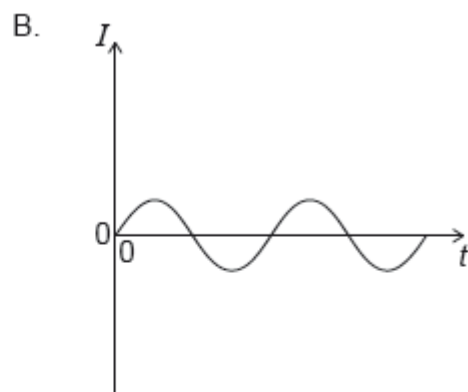
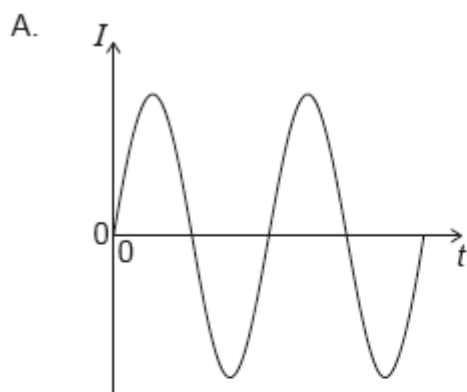
	Peak emf	Periodic time
A.	$2E_0$	$2T$
B.	$2E_0$	$\frac{T}{2}$
C.	E_0	$2T$
D.	E_0	$\frac{T}{2}$

18M.1.HL.TZ1.34

The graph shows the variation with time t of the current I in the primary coil of an ideal transformer.

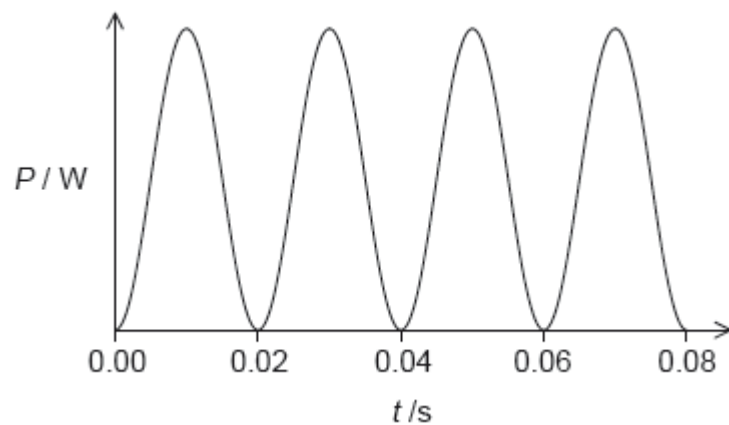


The number of turns in the primary coil is 100 and the number of turns in the secondary coil is 200. Which graph shows the variation with time of the current in the secondary coil?



18M.1.HL.TZ2.35

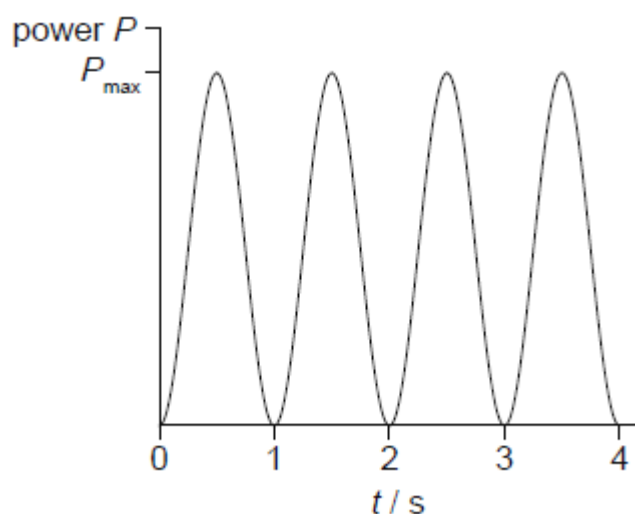
The graph shows the power dissipated in a resistor of $100\ \Omega$ when connected to an alternating current (ac) power supply of root mean square voltage (V_{rms}) $60\ \text{V}$.



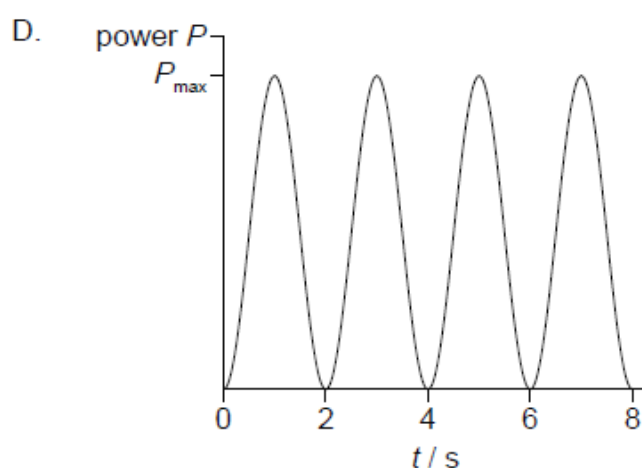
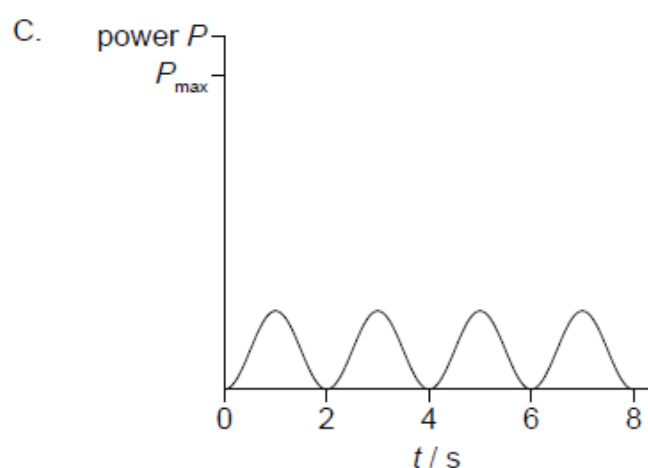
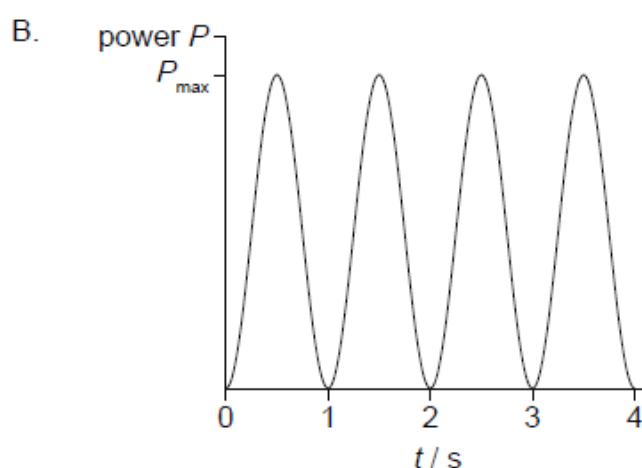
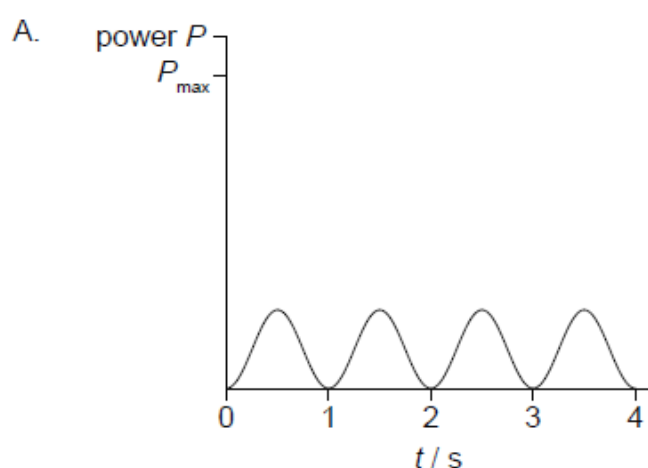
What are the frequency of the ac power supply and the average power dissipated in the resistor?

	Frequency of the ac power supply / Hz	Average power dissipated in the resistor / W
A.	25	36
B.	50	36
C.	25	18
D.	50	18

The graph shows the variation of the peak output power P with time of an alternating current (ac) generator.



Which graph shows the variation of the peak output power with time when the frequency of rotation is decreased?



A resistor designed for use in a direct current (dc) circuit is labelled "50 W, $2\ \Omega$ ". The resistor is connected in series with an alternating current (ac) power supply of peak potential difference 10 V. What is the average power dissipated by the resistor in the ac circuit?

- A. 25 W
- B. 35 W
- C. 50 W
- D. 100 W

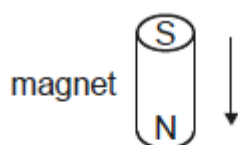
22M.1.HL.TZ1.35

A direct current I in a lamp dissipates power P . What root mean square (rms) value of an alternating current dissipates average power P through the same lamp?

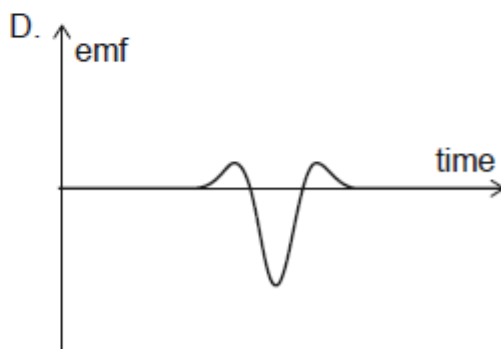
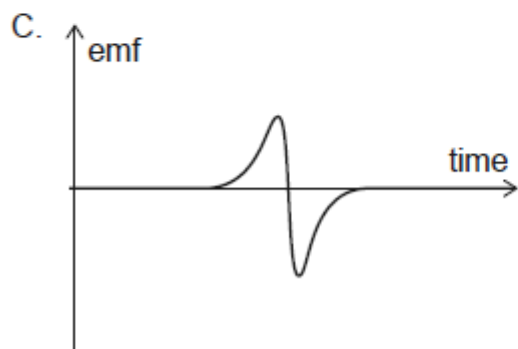
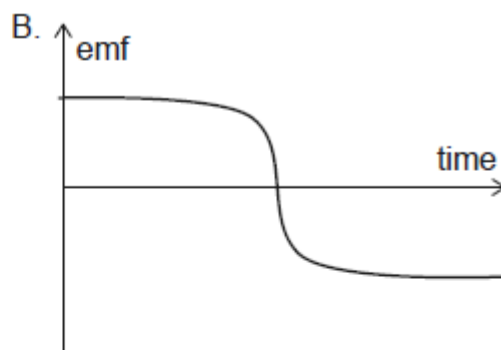
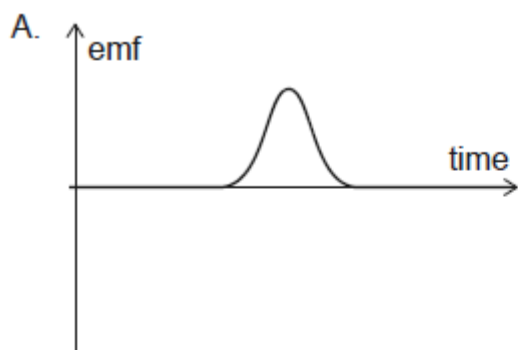
- A. $\frac{I}{2}$
- B. $\frac{I}{\sqrt{2}}$
- C. I
- D. $I\sqrt{2}$

21N.1.HL.TZ0.33

A small magnet is released from rest to drop through a stationary horizontal conducting ring.

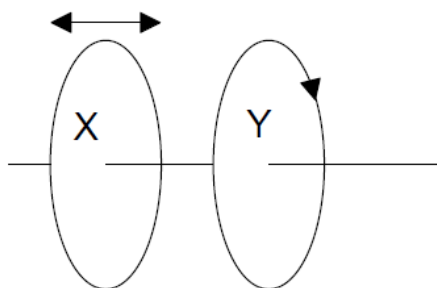


What is the variation with time of the emf induced in the ring?



19N.1.HL.TZ0.33

X and Y are two plane coils parallel to each other that have a common axis. There is a constant direct current in Y.

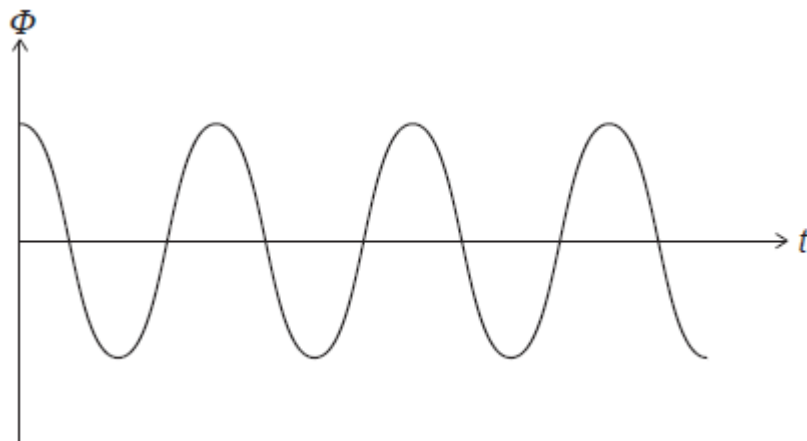


X is first moved towards Y and later is moved away from Y. What, as X moves, is the direction of the current in X relative to that in Y?

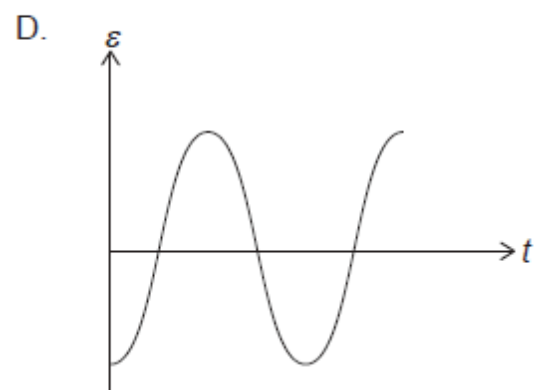
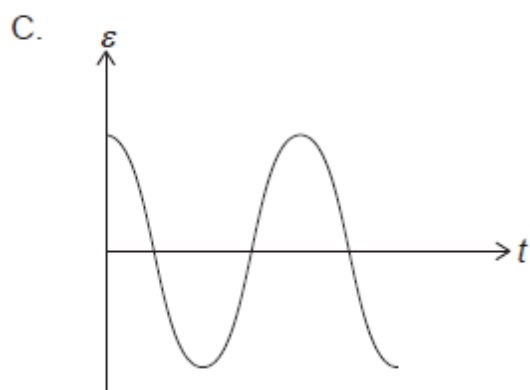
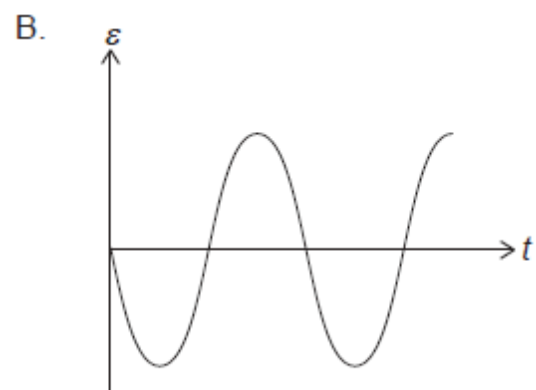
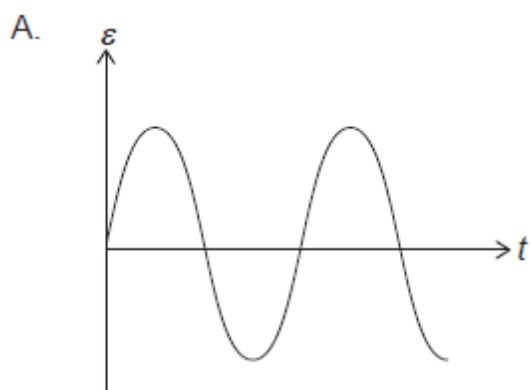
	Current direction in X as X moves towards Y	Current direction in X as X moves away from Y
A.	same direction as in Y	same direction as in Y
B.	same direction as in Y	opposite direction to Y
C.	opposite direction to Y	same direction as in Y
D.	opposite direction to Y	opposite direction to Y

22M.1.HL.TZ1.34

The graph shows the variation of magnetic flux Φ in a coil with time t .



What represents the variation with time of the induced emf ε across the coil?



22M.1.HL.TZ2.35

A conducting bar with vertices PQRS is moving vertically downwards with constant velocity v through a horizontal magnetic field B that is directed into the plane of the page.



Which side of the bar will have the greatest density of electrons?

- A. PQ
- B. QR
- C. RS
- D. SP

21M.1.HL.TZ1.34

The conservation of which quantity explains Lenz's law?

- A. Charge
- B. Energy
- C. Magnetic field
- D. Mass

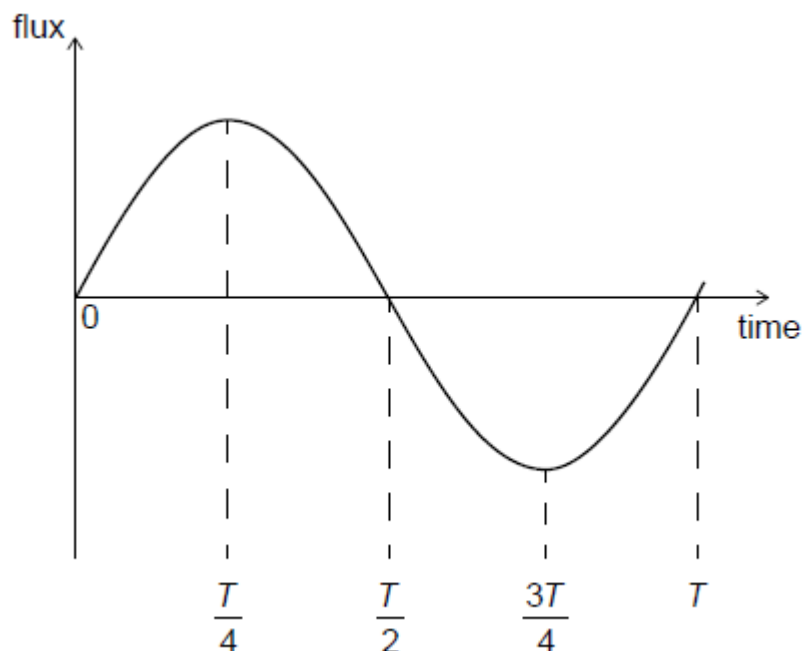
21M.1.HL.TZ1.33

A conducting ring encloses an area of 2.0 cm^2 and is perpendicular to a magnetic field of strength 5.0 mT . The direction of the magnetic field is reversed in a time 4.0 s . What is the average emf induced in the ring?

- A. 0
- B. $0.25 \mu\text{V}$
- C. $0.40 \mu\text{V}$
- D. $0.50 \mu\text{V}$

19M.1.HL.TZ1.35

The graph below shows the variation with time of the magnetic flux through a coil.

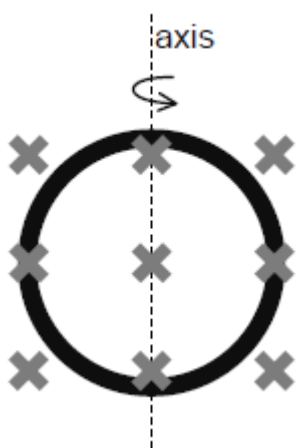


Which of the following gives three times for which the magnitude of the induced emf is a maximum?

- A. $0, \frac{T}{4}, \frac{T}{2}$
- B. $0, \frac{T}{2}, T$
- C. $0, \frac{T}{4}, T$
- D. $\frac{T}{4}, \frac{T}{2}, \frac{3T}{4}$

18N.1.HL.TZ0.33

A ring of area S is in a uniform magnetic field X . Initially the magnetic field is perpendicular to the plane of the ring. The ring is rotated by 180° about the axis in time T .

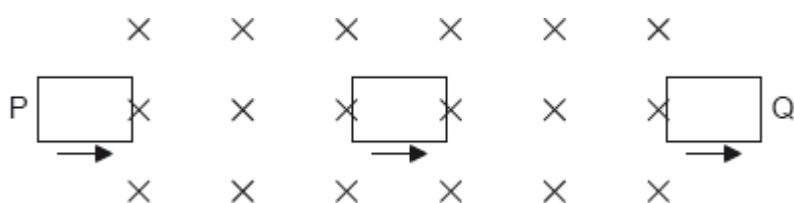


What is the average induced emf in the ring?

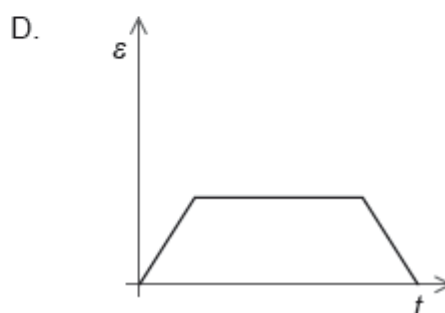
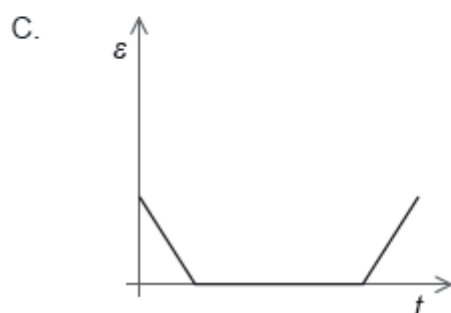
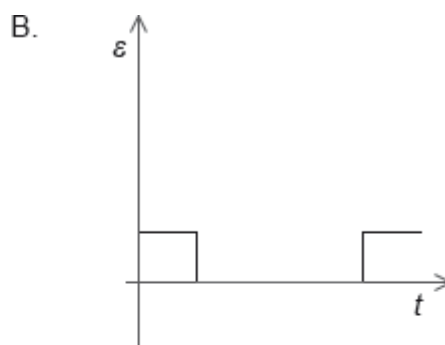
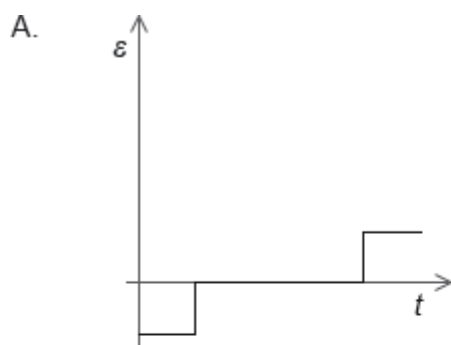
- A. 0
- B. $\frac{XS}{2T}$
- C. $\frac{XS}{T}$
- D. $\frac{2XS}{T}$

18M.1.HL.TZ2.34

A rectangular flat coil moves at constant speed through a uniform magnetic field. The direction of the field is into the plane of the paper.

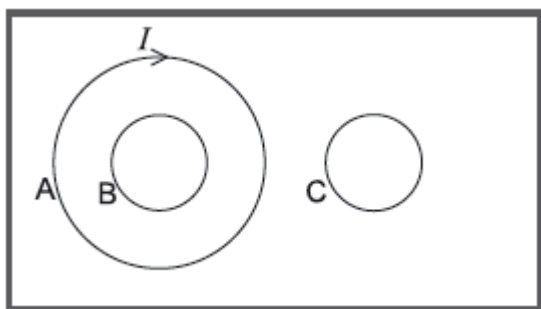


Which graph shows the variation with time t , of the induced emf ε in the coil as it moves from P to Q?



18M.1.HL.TZ2.33

The current I flowing in loop A in a clockwise direction is increasing so as to induce a current both in loops B and C. All three loops are on the same plane.

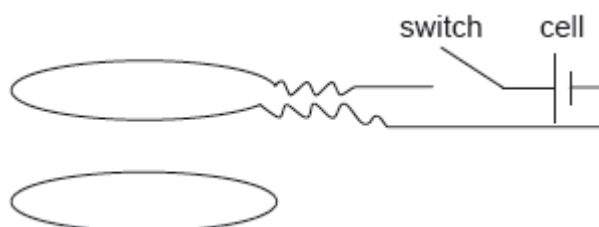


What is the direction of the induced currents in loop B and loop C?

	Loop B	Loop C
A.	clockwise	clockwise
B.	clockwise	anti-clockwise
C.	anti-clockwise	clockwise
D.	anti-clockwise	anti-clockwise

18M.1.HL.TZ1.33

Two identical circular coils are placed one below the other so that their planes are both horizontal. The top coil is connected to a cell and a switch.

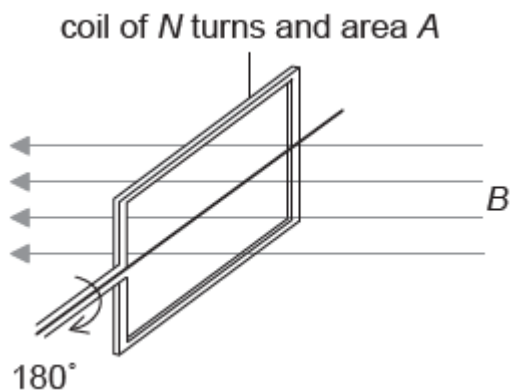


The switch is closed and then opened. What is the force between the coils when the switch is closing and when the switch is opening?

	Switch is closing	Switch is opening
A.	attractive	attractive
B.	attractive	repulsive
C.	repulsive	attractive
D.	repulsive	repulsive

17N.1.HL.TZ0.34

The plane of a coil is positioned at right angles to a magnetic field of flux density B . The coil has N turns, each of area A . The coil is rotated through 180° in time t .

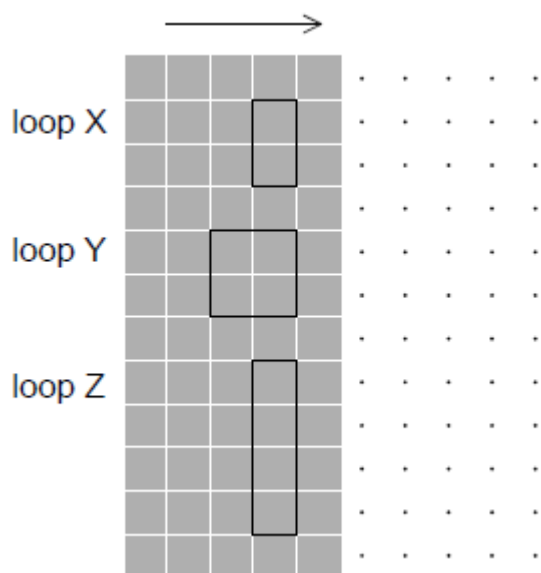


What is the magnitude of the induced emf?

- A. $\frac{BA}{t}$
- B. $\frac{2BA}{t}$
- C. $\frac{BAN}{t}$
- D. $\frac{2BAN}{t}$

17M.1.HL.TZ2.34

Three conducting loops, X, Y and Z, are moving with the same speed from a region of zero magnetic field to a region of uniform non-zero magnetic field.



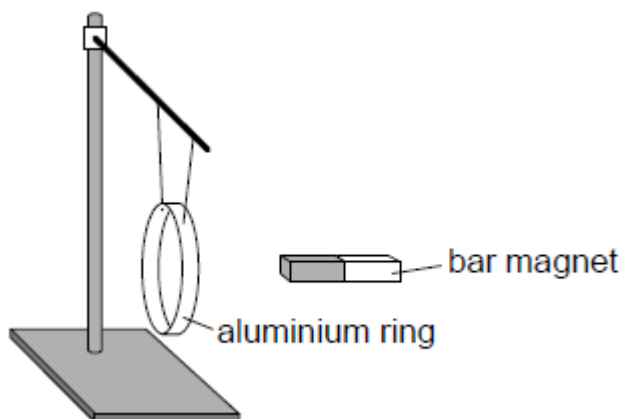
Which loop(s) has/have the largest induced electromotive force (emf) at the instant when the loops enter the magnetic field?

- A. Z only
- B. Y only
- C. Y and Z only

D. X and Y only

17M.1.HL.TZ2.33

The diagram shows a bar magnet near an aluminium ring.



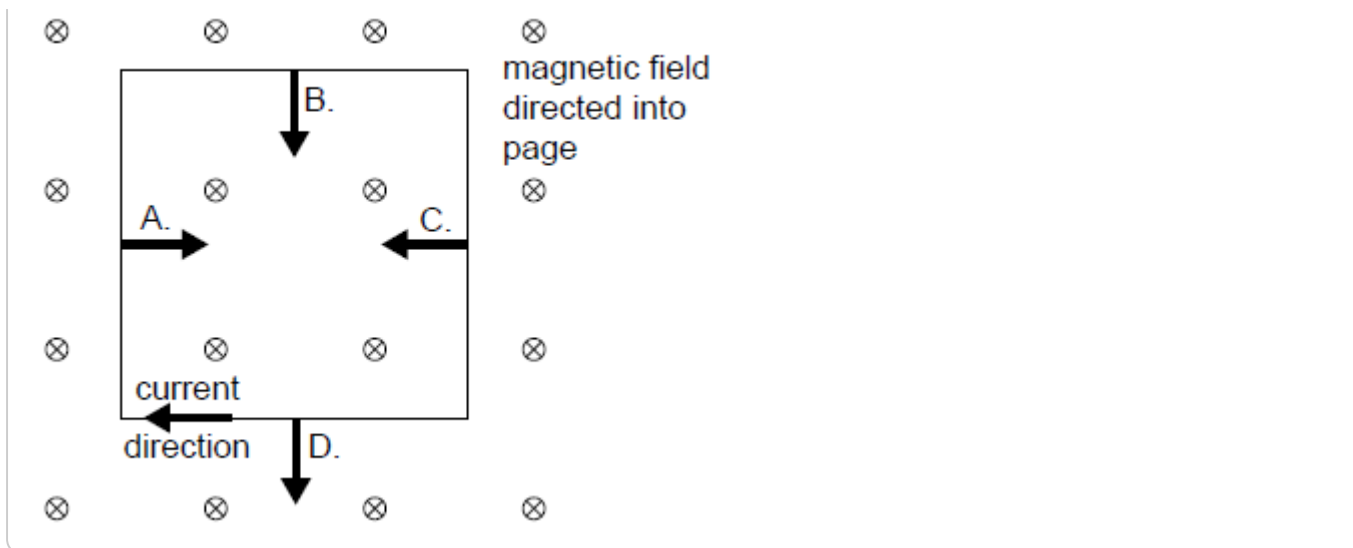
The ring is supported so that it is free to move. The ring is initially at rest. In experiment 1 the magnet is moved towards the ring. In experiment 2 the magnet is moved away from the ring. For each experiment what is the initial direction of motion of the ring?

	Experiment 1	Experiment 2
A.	to the left	to the left
B.	to the left	to the right
C.	to the right	to the left
D.	to the right	to the right

17M.1.HL.TZ1.36

A conducting square coil is placed in a region where there is a uniform magnetic field. The magnetic field is directed into the page. There is a clockwise current in the coil.

What is a correct force that acts on a side of the coil?



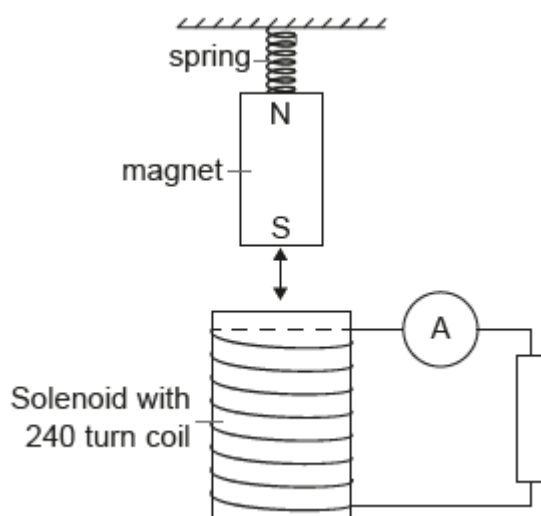
17M.1.HL.TZ1.33

What are the units of magnetic flux and magnetic field strength?

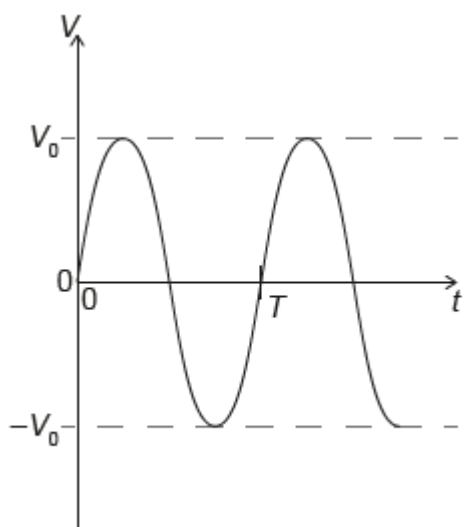
	Magnetic flux	Magnetic field strength
A.	Wb m^{-2}	Wb
B.	Wb	T
C.	Wb	T m^{-2}
D.	T m^{-2}	Wb m^{-2}

21M.1.HL.TZ2.35

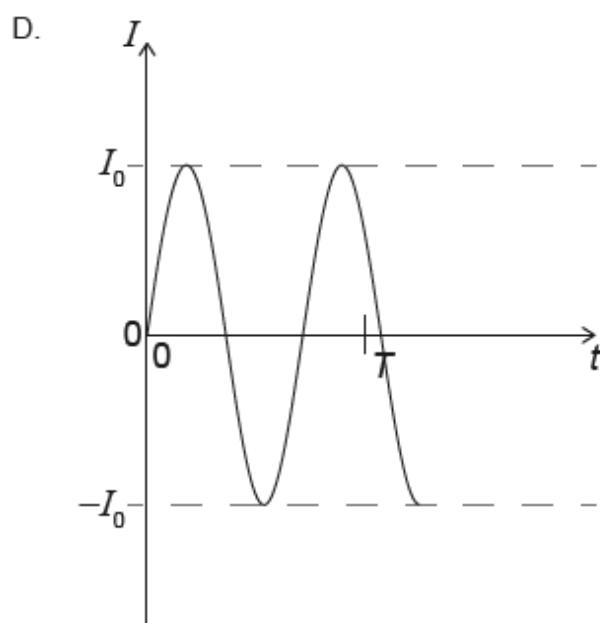
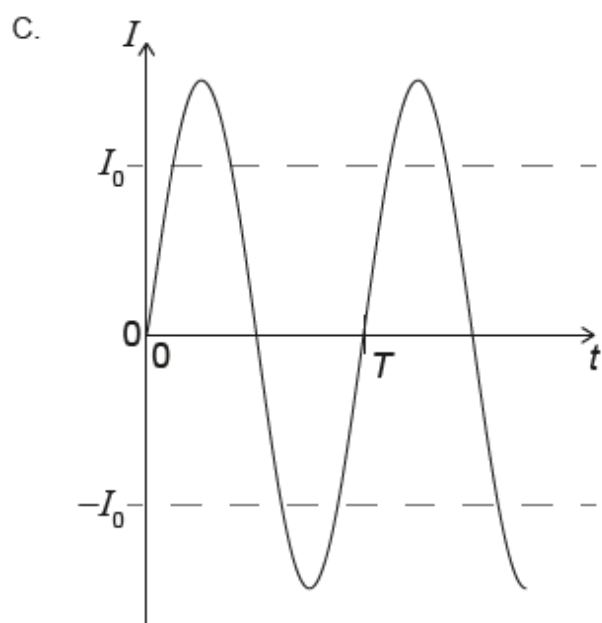
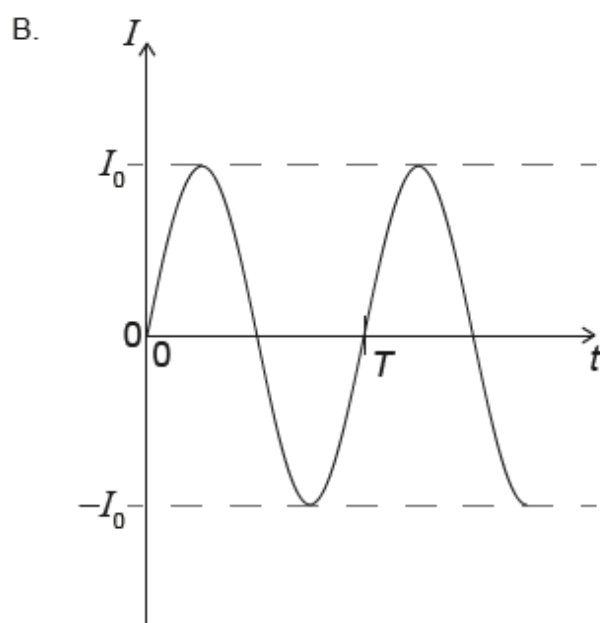
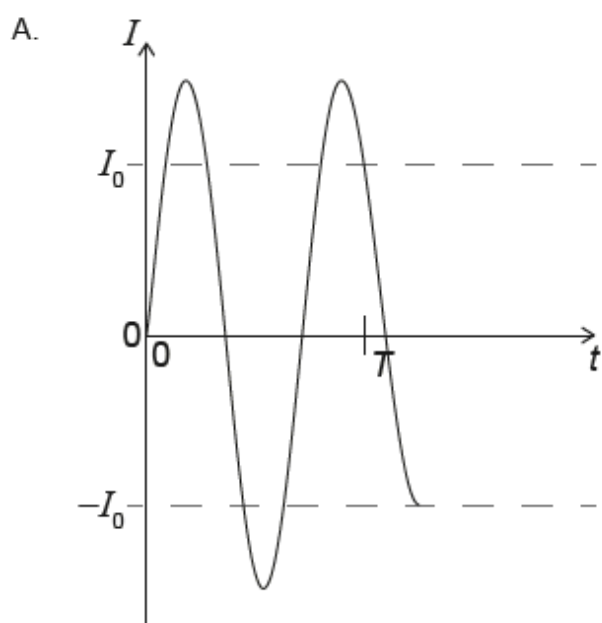
A magnet connected to a spring oscillates above a solenoid with a 240 turn coil as shown.



The graph below shows the variation with time t of the emf across the solenoid with the period, T , of the system shown.



The spring is replaced with one that allows the magnet to oscillate with a higher frequency. Which graph shows the new variation with time t of the current I in the resistor for this new set-up?



21M.1.HL.TZ2.32

A satellite orbits planet X with a speed v_x at a distance r from the centre of planet X. Another satellite orbits planet Y at a speed of v_y at a distance r from the centre of planet Y. The mass of planet X is M and the mass of planet Y is $4M$. What is the ratio of $\frac{v_x}{v_y}$?

- A. 0.25
- B. 0.5
- C. 2.0
- D. 4.0

17M.1.HL.TZ1.30

A satellite at the surface of the Earth has a weight W and gravitational potential energy E_p . The satellite is then placed in a circular orbit with a radius twice that of the Earth.

What is the weight of the satellite and the gravitational potential energy of the satellite when placed in orbit?

	Weight	Gravitational potential energy
A.	$0.25 W$	$0.25 E_p$
B.	$0.5 W$	$0.25 E_p$
C.	$0.25 W$	$0.5 E_p$
D.	$0.5 W$	$0.5 E_p$

16N.1.HL.TZ0.30

What is the unit of $G\epsilon_0$, where G is the gravitational constant and ϵ_0 is the permittivity of free space?

- A. $C \text{ kg}^{-1}$
- B. $C^2 \text{ kg}^{-2}$
- C. $C \text{ kg}$
- D. $C^2 \text{ kg}^2$

16N.1.HL.TZ0.32

A satellite of mass 1500 kg is in the Earth's gravitational field. It moves from a point where the gravitational potential is -30 MJ kg^{-1} to a point where the gravitational potential is -20 MJ kg^{-1} . What is the direction of movement of the satellite and the change in its gravitational potential energy?

	Direction of movement of satellite	Change in gravitational potential energy / GJ
A.	away from Earth	15
B.	away from Earth	75
C.	towards Earth	15
D.	towards Earth	75

17M.1.HL.TZ2.32

Four uniform planets have masses and radii as shown. Which planet has the smallest escape speed?

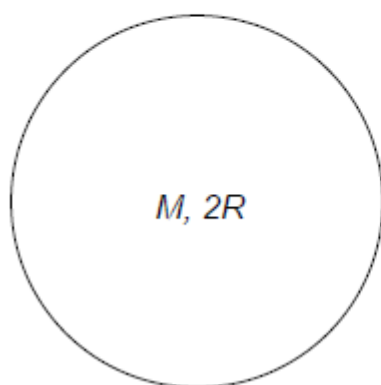
A.



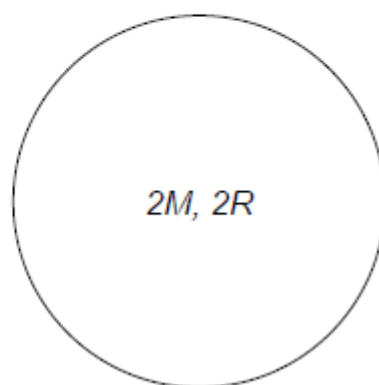
B.



C.

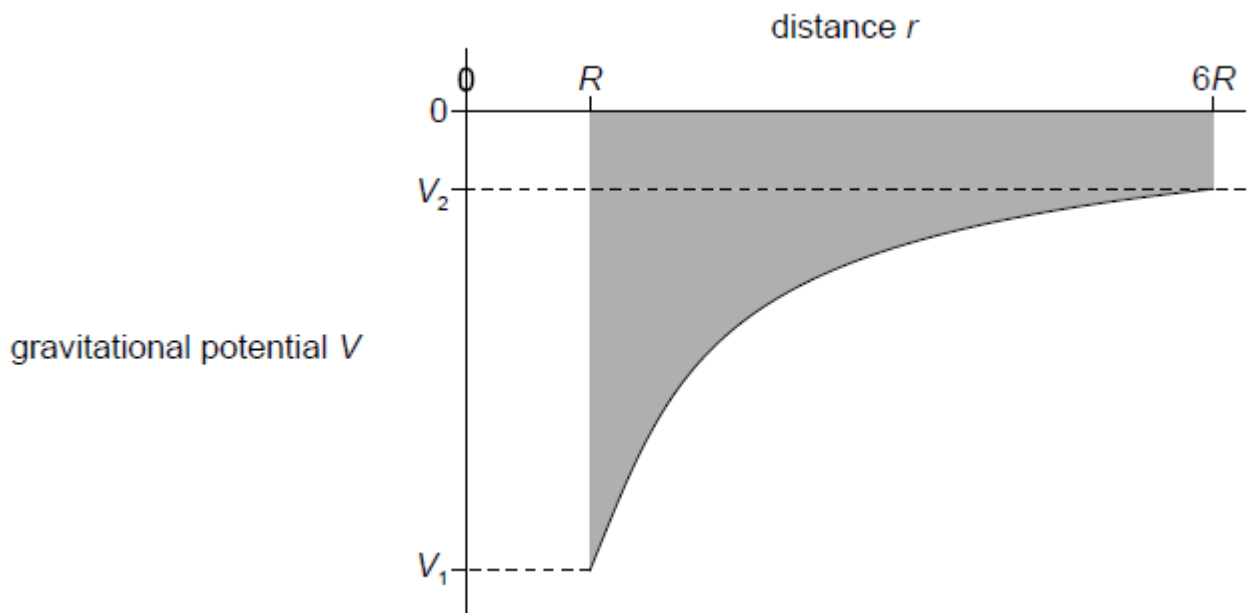


D.



17M.1.HL.TZ2.31

The graph shows the variation of the gravitational potential V with distance r from the centre of a uniform spherical planet. The radius of the planet is R . The shaded area is S .



What is the work done by the gravitational force as a point mass m is moved from the surface of the planet to a distance $6R$ from the centre?

- A. $m(V_2 - V_1)$
- B. $m(V_1 - V_2)$
- C. mS
- D. S

17N.1.HL.TZ0.32

A spacecraft moves towards the Earth under the influence of the gravitational field of the Earth.

The three quantities that depend on the distance r of the spacecraft from the centre of the Earth are the

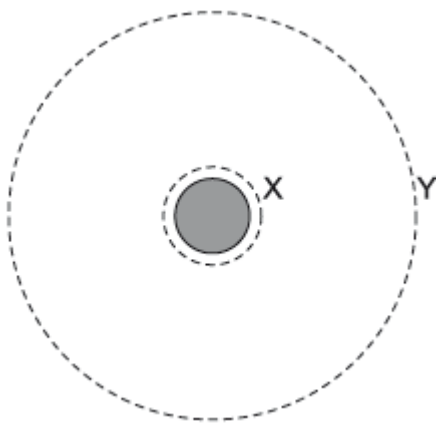
- I. gravitational potential energy of the spacecraft
- II. gravitational field strength acting on the spacecraft
- III. gravitational force acting on the spacecraft.

Which of the quantities are proportional to $\frac{1}{r^2}$?

- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II and III

18M.1.HL.TZ2.31

A satellite orbiting a planet moves from orbit X to orbit Y.



What is the change in the kinetic energy and the change in the gravitational potential energy as a result?

	Kinetic energy	Gravitational potential energy
A.	increases	increases
B.	increases	decreases
C.	decreases	increases
D.	decreases	decreases

18M.1.HL.TZ1.32

An electron of mass m_e orbits an alpha particle of mass m_α in a circular orbit of radius r . Which expression gives the speed of the electron?

- A. $\sqrt{\frac{2ke^2}{m_e r}}$
- B. $\sqrt{\frac{2ke^2}{m_\alpha r}}$
- C. $\sqrt{\frac{4ke^2}{m_e r}}$
- D. $\sqrt{\frac{4ke^2}{m_\alpha r}}$

18M.1.HL.TZ2.32

The mass of the Earth is M_E and the mass of the Moon is M_M . Their respective radii are R_E and R_M .

Which is the ratio $\frac{\text{escape speed from the Earth}}{\text{escape speed from the Moon}}$?

A. $\sqrt{\frac{M_M R_M}{M_E R_E}}$

B. $\sqrt{\frac{M_E R_E}{M_M R_M}}$

C. $\sqrt{\frac{M_E R_M}{M_M R_E}}$

D. $\sqrt{\frac{M_M R_E}{M_E R_M}}$

18N.1.HL.TZ0.31

The gravitational potential at point P due to Earth is V .



What is the definition of the gravitational potential at P?

- A. Work done per unit mass to move a point mass from infinity to P
- B. Work done per unit mass to move a point mass from P to infinity
- C. Work done to move a point mass from infinity to P
- D. Work done to move a point mass from P to infinity

18N.1.HL.TZ0.32

The escape speed for the Earth is v_{esc} . Planet X has half the density of the Earth and twice the radius. What is the escape speed for planet X?

A. $\frac{v_{\text{esc}}}{2}$

B. $\frac{v_{\text{esc}}}{\sqrt{2}}$

C. v_{esc}

D. $\sqrt{2} v_{\text{esc}}$

19M.1.HL.TZ2.10

Satellite X is in orbit around the Earth. An identical satellite Y is in a higher orbit. What is correct for the total energy and the kinetic energy of the satellite Y compared with satellite X?

	Total energy of satellite X	Kinetic energy of satellite X
A.	larger	larger
B.	smaller	larger
C.	larger	smaller
D.	smaller	smaller

19M.1.HL.TZ2.11

The escape speed from a planet of radius R is v_{esc} . A satellite orbits the planet at a distance R from the surface of the planet. What is the orbital speed of the satellite?

- A. $\frac{1}{2}v_{\text{esc}}$
- B. $\frac{\sqrt{2}}{2}v_{\text{esc}}$
- C. $\sqrt{2}v_{\text{esc}}$
- D. $2v_{\text{esc}}$

19M.1.HL.TZ1.33

The escape velocity for an object at the surface of the Earth is v_{esc} . The diameter of the Moon is 4 times smaller than that of the Earth and the mass of the Moon is 81 times smaller than that of the Earth. What is the escape velocity of the object on the Moon?

- A. $\frac{2}{81}v_{\text{esc}}$
- B. $\frac{4}{81}v_{\text{esc}}$
- C. $\frac{2}{9}v_{\text{esc}}$
- D. $\frac{4}{9}v_{\text{esc}}$

19M.1.HL.TZ1.34

A satellite in a circular orbit around the Earth needs to reduce its orbital radius.

What is the work done by the satellite rocket engine and the change in kinetic energy resulting from this shift in orbital height?

	Work done by the satellite rocket engine	Kinetic energy
A.	positive	increase
B.	positive	decrease
C.	negative	increase
D.	negative	decrease

21M.1.HL.TZ1.32

A planet has radius R . The escape speed from the surface of the planet is v . At what distance from the surface of the planet is the orbital speed $0.5v$?

- A. $0.5R$
- B. R
- C. $2R$
- D. $4R$

22M.1.HL.TZ2.32

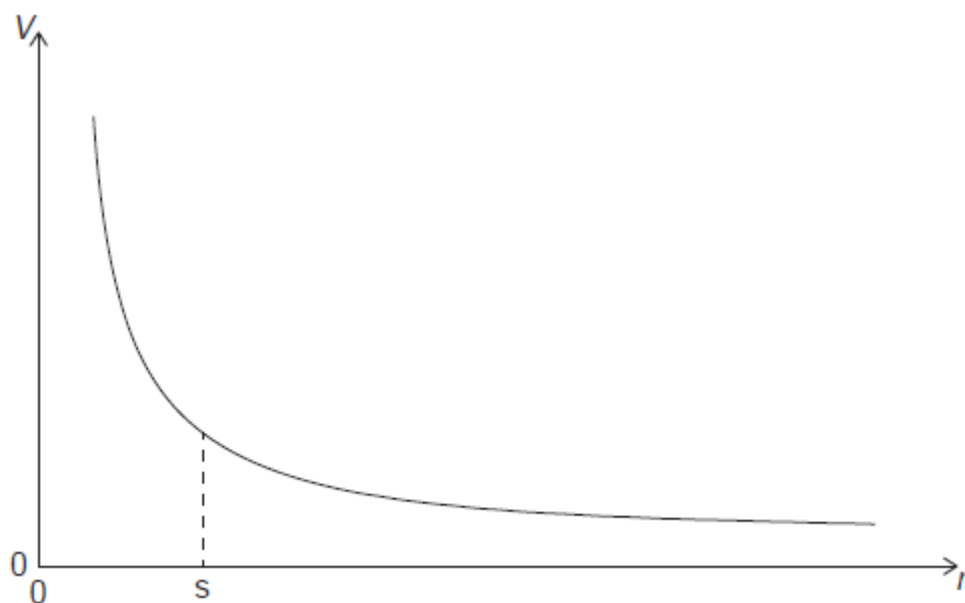
Two positive and two negative charges are located at the corners of a square as shown. Point X is the centre of the square. What is the value of the electric field E and the electric potential V at X due to the four charges?



	Electric field E at X	Electric potential V at X
A.	$E = 0$	$V = 0$
B.	$E \neq 0$	$V = 0$
C.	$E = 0$	$V \neq 0$
D.	$E \neq 0$	$V \neq 0$

22M.1.HL.TZ2.33

The graph shows the variation with distance r of the electric potential V from a charge Q .



What is the electric field strength at distance s ?

- A. The area under the graph between s and infinity
- B. The area under the graph between 0 and s
- C. The gradient of the tangent at s
- D. The negative of the gradient of the tangent at s

22M.1.HL.TZ1.33

An object of mass m is launched from the surface of the Earth. The Earth has a mass M and radius r . The acceleration due to gravity at the surface of the Earth is g . What is the escape speed of the object from the surface of the Earth?

- A. $\sqrt{gr}$
- B. $\sqrt{2gr}$

C. $\sqrt{2Mg\bar{r}}$

D. $\sqrt{2m\bar{g}r}$

21N.1.HL.TZ0.32

A satellite of mass m orbits a planet of mass M in a circular orbit of radius r . What is the work that must be done on the satellite to increase its orbital radius to $2r$?

A. $\frac{GMm}{r}$

B. $\frac{GMm}{2r}$

C. $\frac{GMm}{4r}$

D. $\frac{GMm}{8r}$

19N.1.HL.TZ0.30

The gravitational potential is V at a distance R above the surface of a spherical planet of radius R and uniform density. What is the gravitational potential a distance $2R$ above the surface of the planet?

A. $\frac{V}{4}$

B. $\frac{4V}{9}$

C. $\frac{V}{2}$

D. $\frac{2V}{3}$

19N.1.HL.TZ0.32

An electron enters a uniform electric field of strength E with a velocity v . The direction of v is not parallel to E . What is the path of the electron after entering the field?

A. Circular

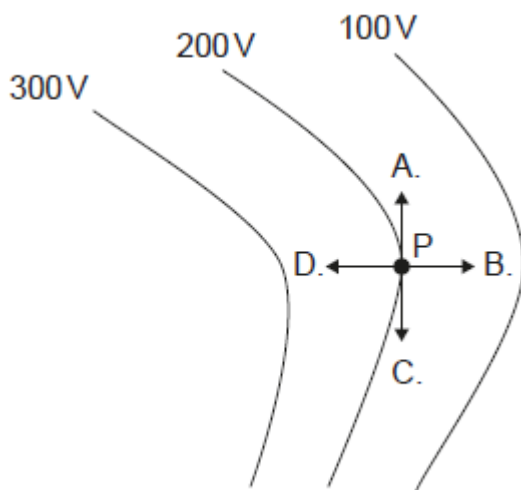
B. Parabolic

C. Parallel to E

D. Parallel to v

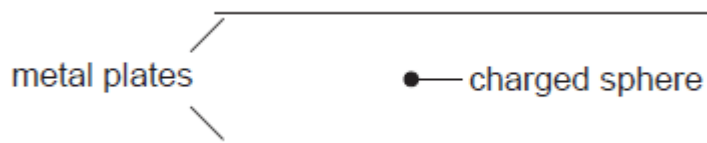
21N.1.HL.TZ0.30

The diagram shows equipotential lines for an electric field. Which arrow represents the acceleration of an electron at point P?



22M.1.HL.TZ1.32

A charged sphere in a gravitational field is initially stationary between two parallel metal plates. There is a potential difference V between the plates.



Three changes can be made:

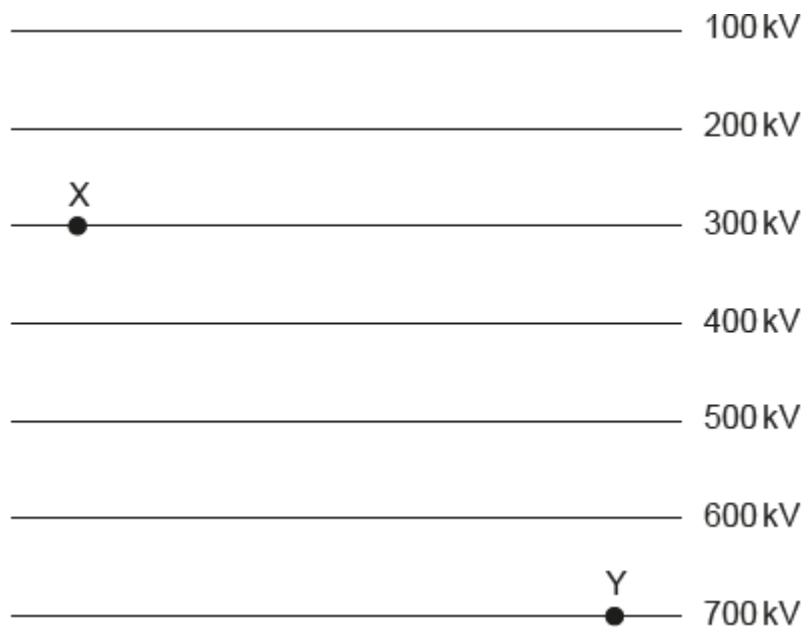
- I. Increase the separation of the metal plates
- II. Increase V
- III. Apply a magnetic field into the plane of the paper

What changes made separately will cause the charged sphere to accelerate?

- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II and III

21M.1.HL.TZ1.30

A particle with charge $-2.5 \times 10^{-6} \text{ C}$ moves from point X to point Y due to a uniform electrostatic field. The diagram shows some equipotential lines of the field.



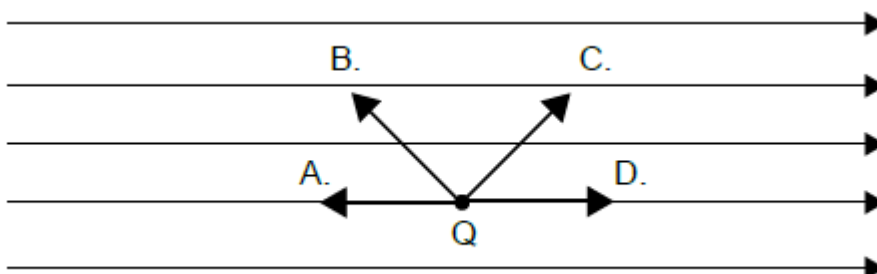
What is correct about the motion of the particle from X to Y and the magnitude of the work done by the field on the particle?

	Motion of the particle from X to Y	Magnitude of the work done by the field on the particle
A.	uniform linear	0 J
B.	uniform linear	1 J
C.	uniformly accelerated	0 J
D.	uniformly accelerated	1 J

19M.1.HL.TZ1.32

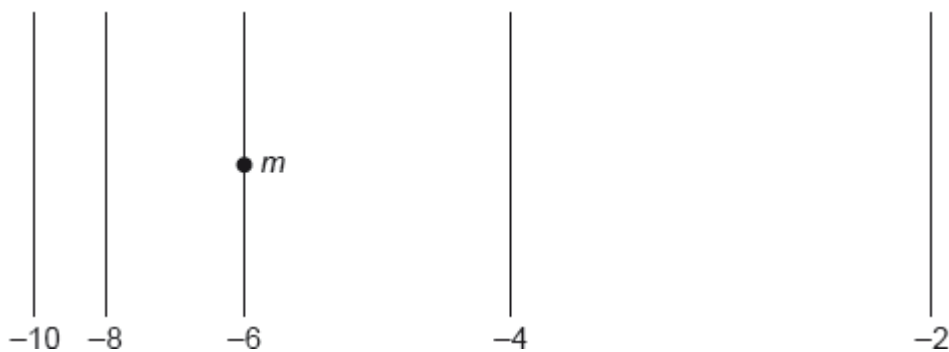
A negative charge Q is to be moved within an electric field E , to equidistant points from its position, as shown.

Which path requires the most work done?



18M.1.HL.TZ1.31

The diagram shows 5 gravitational equipotential lines. The gravitational potential on each line is indicated. A point mass m is placed on the middle line and is then released. Values given in MJ kg^{-1} .

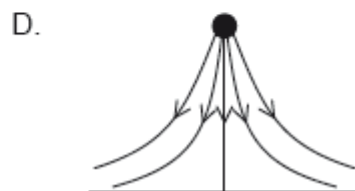
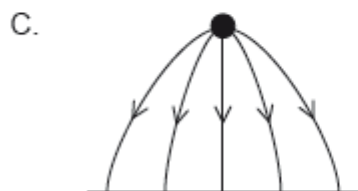


Which is correct about the direction of motion and the acceleration of the point mass?

	Direction	Acceleration
A.	to the right	decreasing
B.	to the right	increasing
C.	to the left	decreasing
D.	to the left	increasing

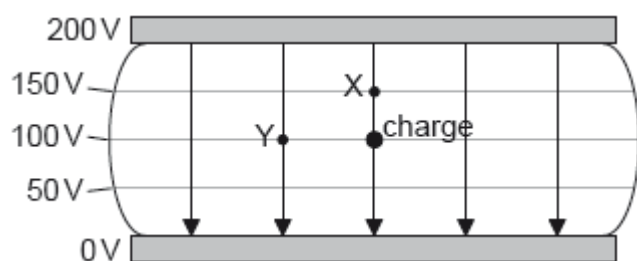
18M.1.HL.TZ2.30

A positive point charge is placed above a metal plate at zero electric potential. Which diagram shows the pattern of electric field lines between the charge and the plate?



18M.1.HL.TZ2.29

The diagram shows the electric field and the electric equipotential surfaces between two charged parallel plates. The potential difference between the plates is 200 V.

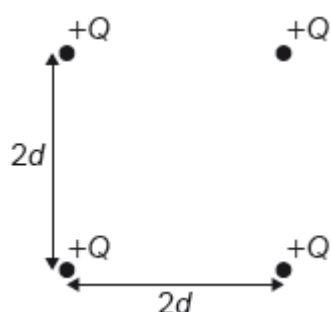


What is the work done, in nJ, by the electric field in moving a negative charge of magnitude 1 nC from the position shown to X and to Y?

	To X	To Y
A.	50	0
B.	-50	0
C.	50	100
D.	-50	-100

18M.1.HL.TZ1.30

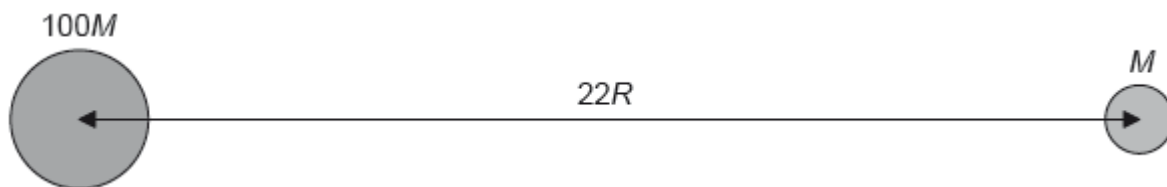
Four identical, positive, point charges of magnitude Q are placed at the vertices of a square of side $2d$. What is the electric potential produced at the centre of the square by the four charges?



- A. 0
- B. $\frac{4kQ}{d}$
- C. $\frac{\sqrt{2}kQ}{d}$
- D. $\frac{2\sqrt{2}kQ}{d}$

18M.1.HL.TZ2.28

A moon of mass M orbits a planet of mass $100M$. The radius of the planet is R and the distance between the centres of the planet and moon is $22R$.



What is the distance from the centre of the planet at which the total gravitational potential has a maximum value?

- A. $2R$
- B. $11R$
- C. $20R$
- D. $2R$ and $20R$

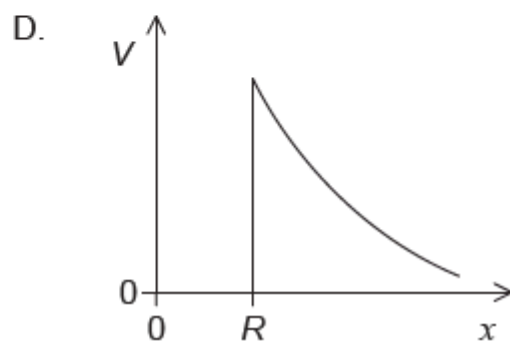
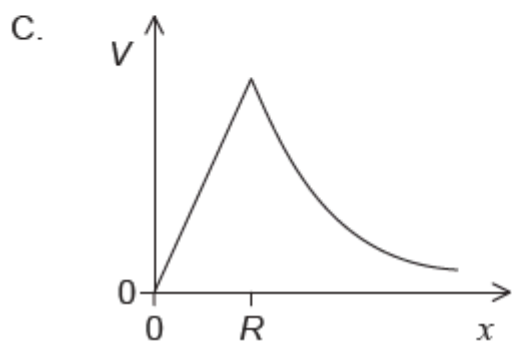
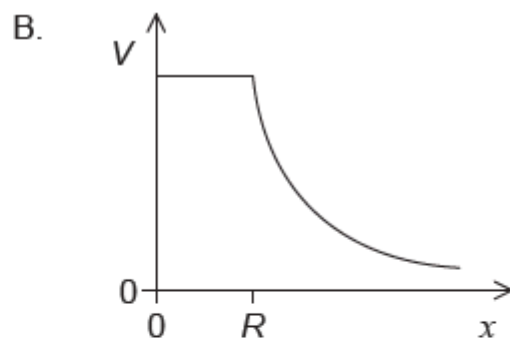
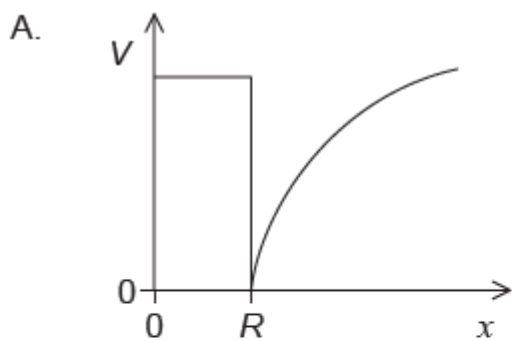
17N.1.HL.TZ0.31

A charge of -3 C is moved from A to B and then back to A. The electric potential at A is $+10\text{ V}$ and the electric potential at B is -20 V . What is the work done in moving the charge from A to B and the total work done?

	Work done in moving from A to B / J	Total work done / J
A.	30	0
B.	30	60
C.	90	0
D.	90	180

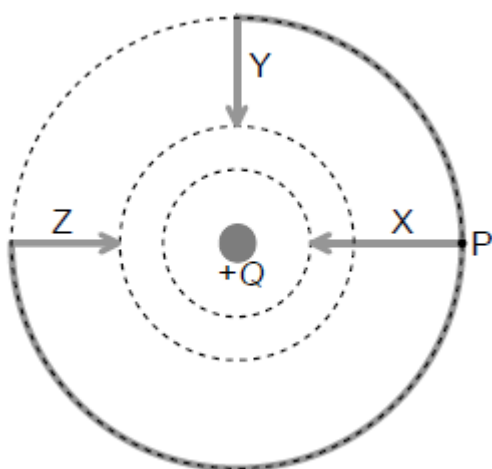
17N.1.HL.TZ0.33

An isolated hollow metal sphere of radius R carries a positive charge. Which graph shows the variation of potential V with distance x from the centre of the sphere?



17M.1.HL.TZ2.30

A positive charge Q is deposited on the surface of a small sphere. The dotted lines represent equipotentials.



A small positive point charge is moved from point P closer to the sphere along three different paths X, Y and Z. The work done along each path is W_X , W_Y and W_Z . What is a correct comparison of W_X , W_Y and W_Z ?

- A. $W_Z > W_Y > W_X$
- B. $W_X > W_Y = W_Z$
- C. $W_X = W_Y = W_Z$
- D. $W_Z = W_Y > W_X$

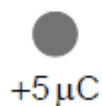
17M.1.HL.TZ1.31

Two point charges are at rest as shown.

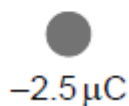
At which position is the electric field strength greatest?

C.

A.



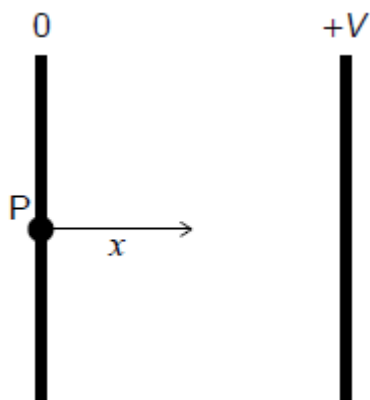
B.



D.

17M.1.HL.TZ1.29

An electric field acts in the space between two charged parallel plates. One plate is at zero potential and the other is at potential $+V$.



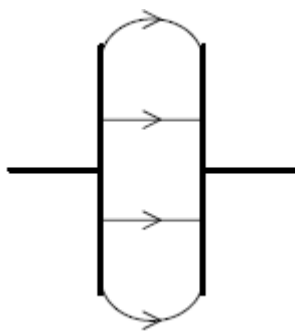
The distance x is measured from point P in the direction perpendicular to the plate.

What is the dependence of the electric field strength E on x and what is the dependence of the electric potential V on x ?

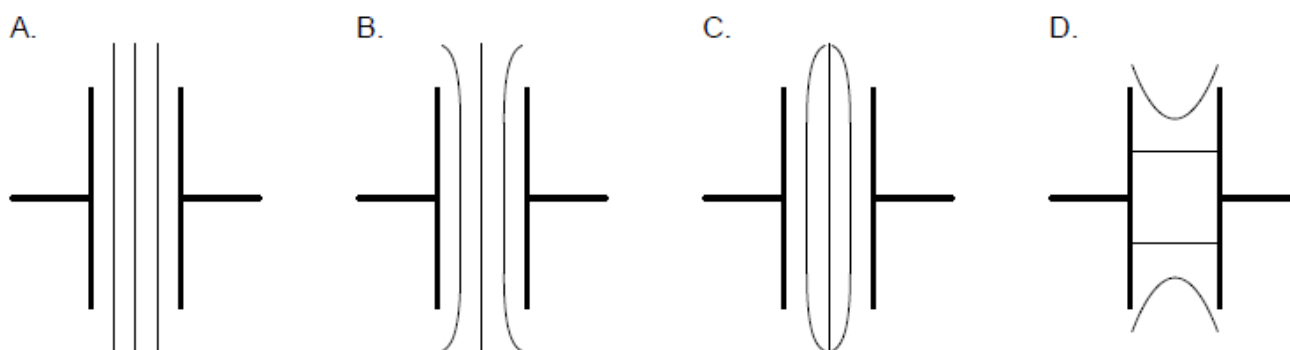
	E	V
A.	proportional to $\frac{1}{x^2}$	constant
B.	constant	proportional to x
C.	proportional to x	proportional to x
D.	proportional to x^2	constant

16N.1.HL.TZ0.31

Two parallel metal plates are connected to a dc power supply. An electric field forms in the space between the plates as shown.

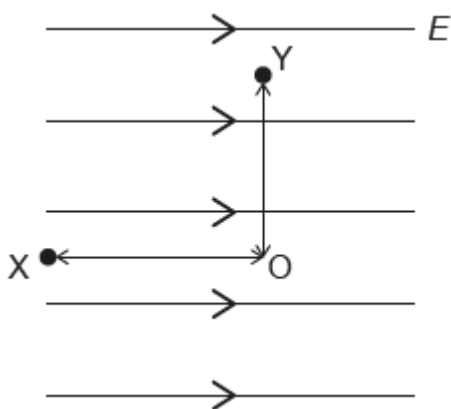


What is the shape of the equipotential surfaces that result from this arrangement?



21M.1.HL.TZ2.31

The points X and Y are in a uniform electric field of strength E . The distance OX is x and the distance OY is y .



What is the magnitude of the change in electric potential between X and Y?

- A. Ex
- B. Ey
- C. $E(x + y)$
- D. $E\sqrt{x^2 + y^2}$

21M.1.HL.TZ2.30

An object of mass m released from rest near the surface of a planet has an initial acceleration z . What is the gravitational field strength near the surface of the planet?

- A. z
- B. $\frac{z}{m}$
- C. mz
- D. $\frac{m}{z}$

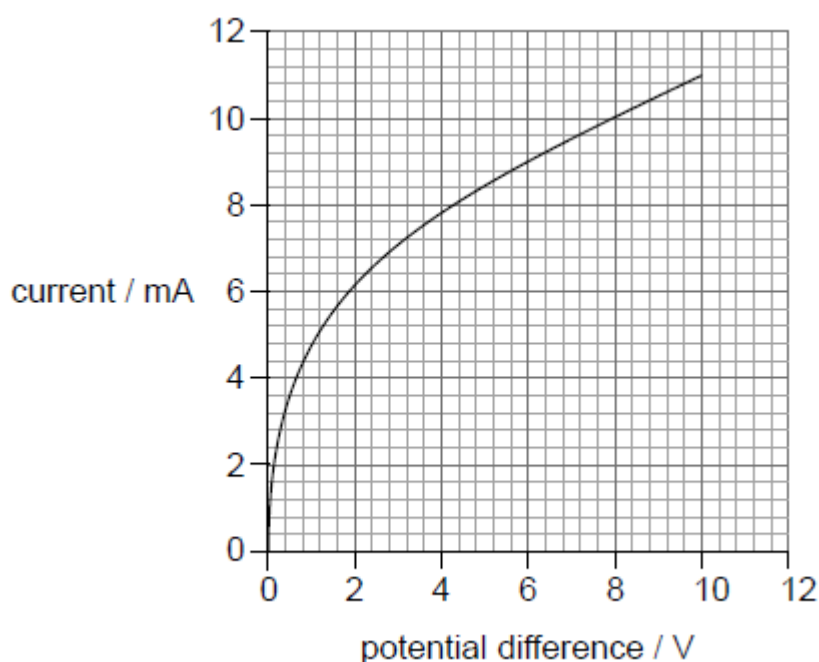
17M.1.SL.TZ1.19

An electron is accelerated through a potential difference of 2.5 MV. What is the change in kinetic energy of the electron?

- A. $0.4\mu\text{J}$
- B. 0.4 nJ
- C. 0.4 pJ
- D. 0.4 fJ

17M.1.SL.TZ1.18

The graph shows the variation of current with potential difference for a filament lamp.



What is the resistance of the filament when the potential difference across it is 6.0 V?

- A. 0.5 m Ω
- B. 1.5 m Ω
- C. 670 Ω
- D. 2000 Ω

17M.1.SL.TZ1.20

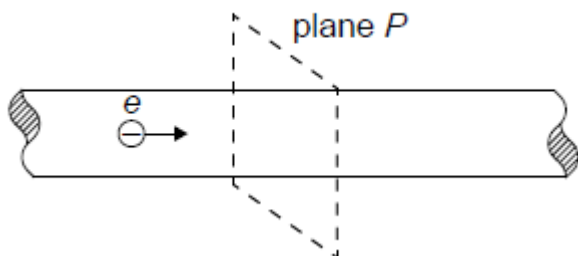
A cell is connected in series with a resistor and supplies a current of 4.0 A for a time of 500 s. During this time, 1.5 kJ of energy is dissipated in the cell and 2.5 kJ of energy is dissipated in the resistor.

What is the emf of the cell?

- A. 0.50 V
- B. 0.75 V
- C. 1.5 V
- D. 2.0 V

17M.1.HL.TZ1.17

Electrons, each with a charge e , move with speed v along a metal wire. The electric current in the wire is I .

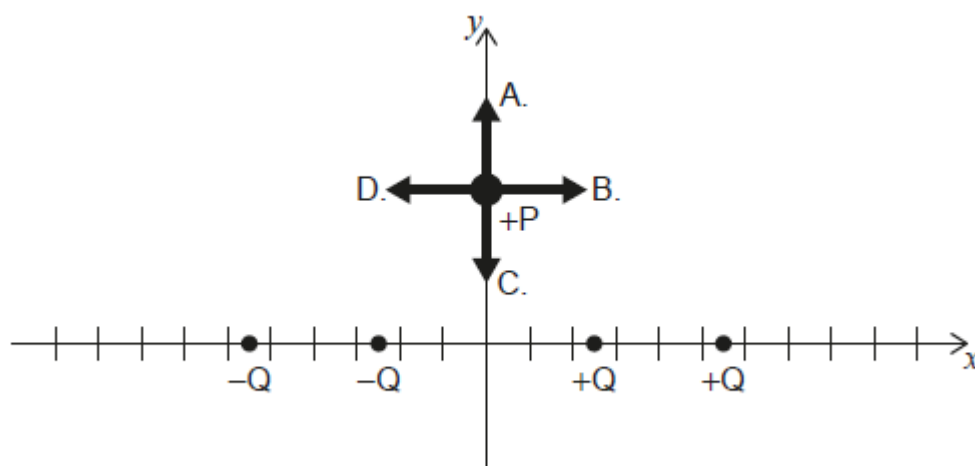


Plane P is perpendicular to the wire. How many electrons pass through plane P in each second?

- A. $\frac{e}{I}$
- B. $\frac{ve}{I}$
- C. $\frac{I}{ve}$
- D. $\frac{I}{e}$

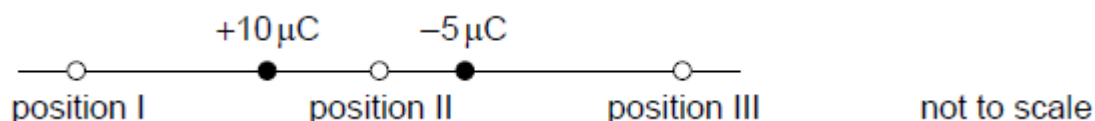
22M.1.HL.TZ2.16

Four particles, two of charge $+Q$ and two of charge $-Q$, are positioned on the x -axis as shown. A particle P with a positive charge is placed on the y -axis. What is the direction of the net electrostatic force on this particle?



16N.1.SL.TZ0.18

A $-5\mu\text{C}$ charge and a $+10\mu\text{C}$ charge are a fixed distance apart.

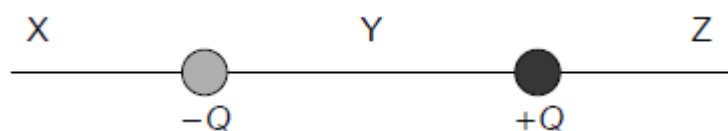


Where can the electric field be zero?

- A. position I only
- B. position II only
- C. position III only
- D. positions I, II and III

17M.1.SL.TZ2.18

The diagram shows two equal and opposite charges that are fixed in place.



At which points is the net electric field directed to the right?

- A. X and Y only
- B. Z and Y only
- C. X and Z only
- D. X, Y and Z

17M.1.SL.TZ2.19

A wire has variable cross-sectional area. The cross-sectional area at Y is double that at X.



At X, the current in the wire is I and the electron drift speed is v . What is the current and the electron drift speed at Y?

	Current	Drift speed
A.	I	v
B.	I	$\frac{v}{2}$
C.	$2I$	v
D.	$2I$	$\frac{v}{2}$

17N.1.HL.TZ0.15

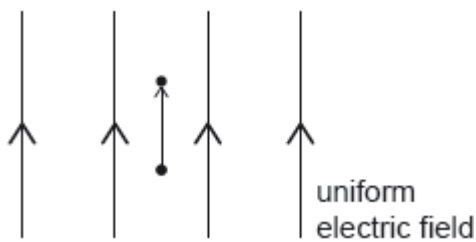
Two wires, X and Y, are made from the same metal. The wires are connected in series. The radius of X is twice that of Y. The carrier drift speed in X is v_X and in Y it is v_Y .

What is the value of the ratio $\frac{v_X}{v_Y}$?

- A. 0.25
- B. 0.50
- C. 2.00
- D. 4.00

18M.1.HL.TZ1.15

An ion of charge $+Q$ moves vertically upwards through a small distance s in a uniform vertical electric field. The electric field has a strength E and its direction is shown in the diagram.

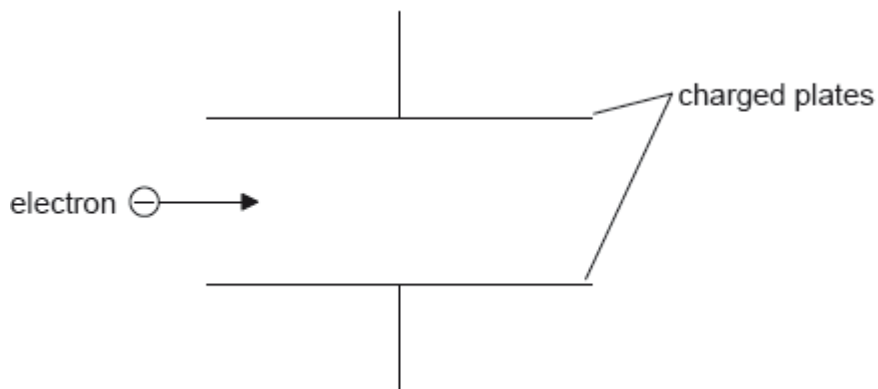


What is the electric potential difference between the initial and final position of the ion?

- A. $\frac{EQ}{s}$
- B. EQs
- C. Es
- D. $\frac{E}{s}$

18M.1.SL.TZ2.20

An electron enters the region between two charged parallel plates initially moving parallel to the plates.



The electromagnetic force acting on the electron

- A. causes the electron to decrease its horizontal speed.
- B. causes the electron to increase its horizontal speed.
- C. is parallel to the field lines and in the opposite direction to them.
- D. is perpendicular to the field direction.

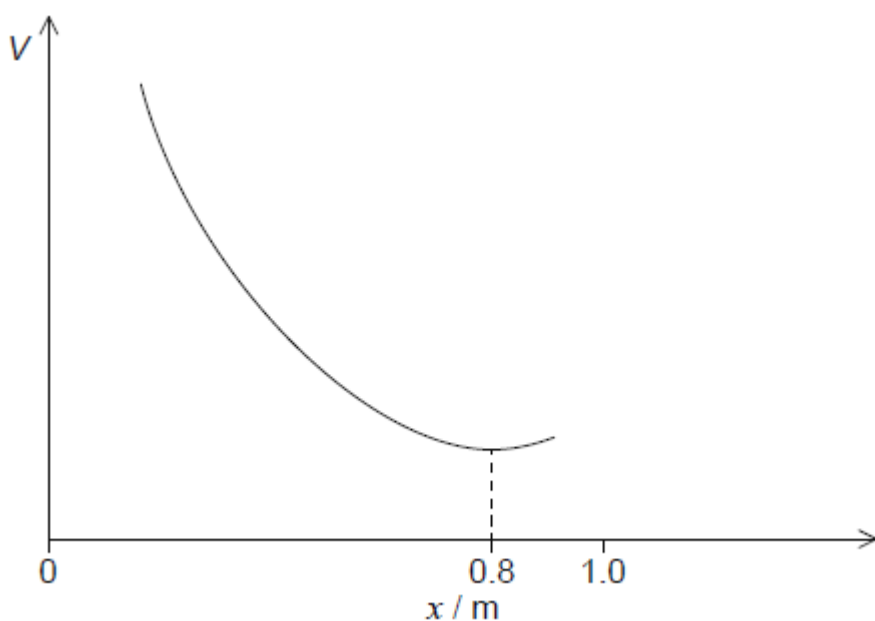
18N.1.SL.TZ0.18

Two copper wires X and Y are connected in series. The diameter of Y is double that of X. The drift speed in X is v . What is the drift speed in Y?

- A. $\frac{v}{4}$
- B. $\frac{v}{2}$
- C. $2v$
- D. $4v$

18N.1.HL.TZ0.30

Two point charges Q_1 and Q_2 are one metre apart. The graph shows the variation of electric potential V with distance x from Q_1 .



What is $\frac{Q_1}{Q_2}$?

- A. $\frac{1}{16}$
- B. $\frac{1}{4}$
- C. 4
- D. 16

19M.1.SL.TZ1.22

Charge flows through a liquid. The charge flow is made up of positive and negative ions. In one second 0.10 C of negative ions flow in one direction and 0.10 C of

positive ions flow in the opposite direction.

What is the magnitude of the electric current flowing through the liquid?

- A. 0 A
- B. 0.05 A
- C. 0.10 A
- D. 0.20 A

21M.1.SL.TZ1.18

Two charges Q_1 and Q_2 , each equal to 2 nC, are separated by a distance 3 m in a vacuum. What is the electric force on Q_2 and the electric field due to Q_1 at the position of Q_2 ?

	Electric force on Q_2	Electric field due to Q_1 at the position of Q_2
A.	4×10^{-9} N	2 N C^{-1}
B.	4 N	2 N C^{-1}
C.	4×10^{-9} N	$2 \times 10^{-9} \text{ N C}^{-1}$
D.	4 N	$2 \times 10^{-9} \text{ N C}^{-1}$

19M.1.SL.TZ2.2

What is the unit of electrical potential difference expressed in fundamental SI units?

- A. $\text{kg m s}^{-1} \text{ C}^{-1}$
- B. $\text{kg m}^2 \text{ s}^{-2} \text{ C}^{-1}$
- C. $\text{kg m}^2 \text{ s}^{-3} \text{ A}^{-1}$
- D. $\text{kg m}^2 \text{ s}^{-1} \text{ A}$

19M.1.HL.TZ1.16

Two parallel plates are a distance apart with a potential difference between them. A point charge moves from the negatively charged plate to the positively charged plate. The charge gains kinetic energy W . The distance between the plates is doubled and the potential difference between them is halved. What is the kinetic energy gained by an identical charge moving between these plates?

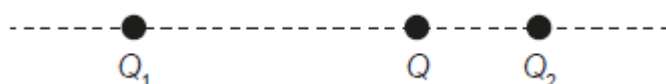
- A. $\frac{W}{2}$

- B. W
- C. $2W$
- D. $4W$

22M.1.SL.TZ2.19

A charge Q is at a point between two electric charges Q_1 and Q_2 . The net electric force on Q is zero. Charge Q_1 is further from Q than charge Q_2 .

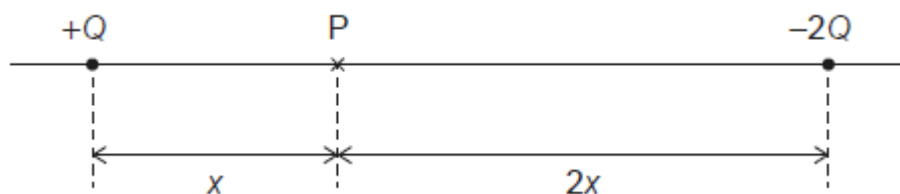
What is true about the signs of the charges Q_1 and Q_2 and their magnitudes?



	Signs of charges Q_1 and Q_2	Magnitudes
A.	same	$Q_1 > Q_2$
B.	same	$Q_1 < Q_2$
C.	opposite	$Q_1 > Q_2$
D.	opposite	$Q_1 < Q_2$

21N.1.SL.TZ0.18

A charge $+Q$ and a charge $-2Q$ are a distance $3x$ apart. Point P is on the line joining the charges, at a distance x from $+Q$.



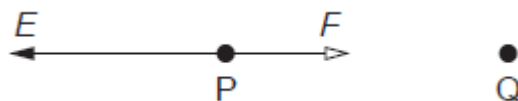
The magnitude of the electric field produced at P by the charge $+Q$ alone is E .

What is the total electric field at P?

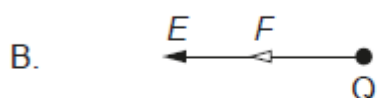
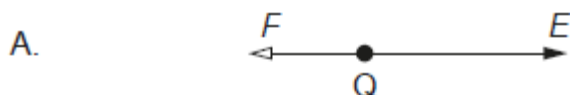
- A. $\frac{E}{2}$ to the right
- B. $\frac{E}{2}$ to the left
- C. $\frac{3E}{2}$ to the right
- D. $\frac{3E}{2}$ to the left

22M.1.SL.TZ1.19

P and Q are two opposite point charges. The force F acting on P due to Q and the electric field strength E at P are shown.

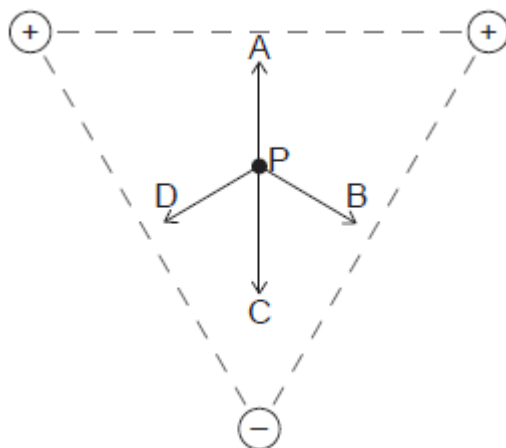


Which diagram shows the force on Q due to P and the electric field strength at Q?



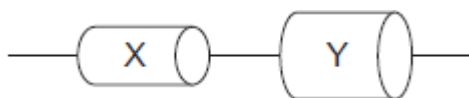
22M.1.SL.TZ1.20

Three point charges of equal magnitude are placed at the vertices of an equilateral triangle. The signs of the charges are shown. Point P is equidistant from the vertices of the triangle. What is the direction of the resultant electric field at P?



22M.1.SL.TZ1.22

Two cylinders, X and Y, made from the same material, are connected in series.



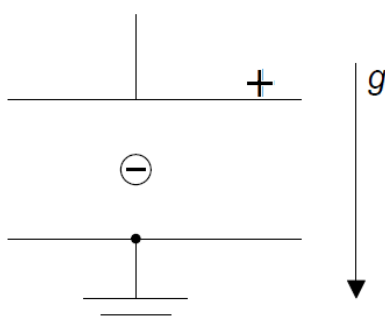
The cross-sectional area of Y is twice that of X. The drift speed of the electrons in X is v_X and in Y it is v_Y .

What is the ratio $\frac{v_X}{v_Y}$?

- A. 4
- B. 2
- C. 1
- D. $\frac{1}{2}$

19N.1.SL.TZ0.17

A negatively charged particle in a uniform gravitational field is positioned mid-way between two charged conducting plates.



The potential difference between the plates is adjusted until the particle is held at rest relative to the plates.

What change will cause the particle to accelerate downwards relative to the plates?

- A. Decreasing the charge on the particle
- B. Decreasing the separation of the plates
- C. Increasing the length of the plates
- D. Increasing the potential difference between the plates

19N.1.SL.TZ0.18

A thin copper wire and a thick copper wire are connected in series to an electric cell. Which quantity will be greater in the thin wire?

- A. Current
- B. Number of free charge carriers per unit volume
- C. Net number of charge carriers crossing a section of a wire every second
- D. Drift speed of the charge carriers

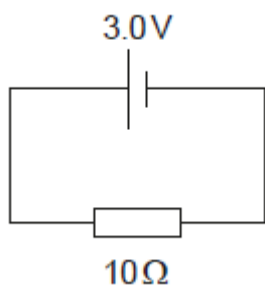
19N.1.HL.TZ0.31

The force acting between two point charges is F when the separation of the charges is x . What is the force between the charges when the separation is increased to $3x$?

- A. $\frac{F}{3}$
- B. $\frac{F}{3x^2}$
- C. $\frac{F}{9}$
- D. $\frac{F}{9x^2}$

21N.1.HL.TZ0.17

A cell has an emf of 3.0 V and an internal resistance of $2.0\ \Omega$. The cell is connected in series with a resistance of $10\ \Omega$.

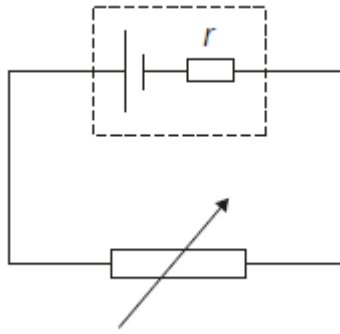


What is the terminal potential difference of the cell?

- A. 0.5 V
- B. 1.5 V
- C. 2.5 V
- D. 3.0 V

21N.1.SL.TZ0.21

A variable resistor is connected in series to a cell with internal resistance r as shown.



The resistance of the variable resistor is increased. What happens to the power dissipated in the cell and to the terminal potential difference of the cell?

	Power dissipated in the cell	Terminal potential difference of the cell
A.	decreases	increases
B.	increases	increases
C.	decreases	decreases
D.	increases	decreases

21M.1.SL.TZ1.20

For a real cell in a circuit, the terminal potential difference is at its closest to the emf when

- A. the internal resistance is much smaller than the load resistance.
- B. a large current flows in the circuit.
- C. the cell is not completely discharged.
- D. the cell is being recharged.

18M.1.HL.TZ1.17

When an electric cell of negligible internal resistance is connected to a resistor of resistance $4R$, the power dissipated in the resistor is P .

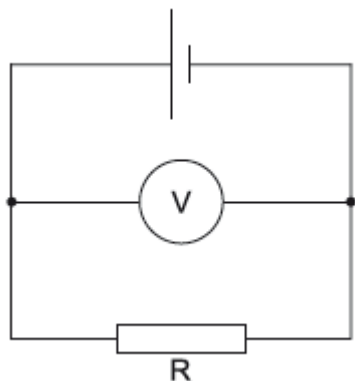
What is the power dissipated in a resistor of resistance value R when it is connected to the same cell?

- A. $\frac{P}{4}$
- B. P

- C. $4P$
- D. $16P$

18M.1.SL.TZ2.22

A cell has an emf of 4.0 V and an internal resistance of $2.0\ \Omega$. The ideal voltmeter reads 3.2 V.

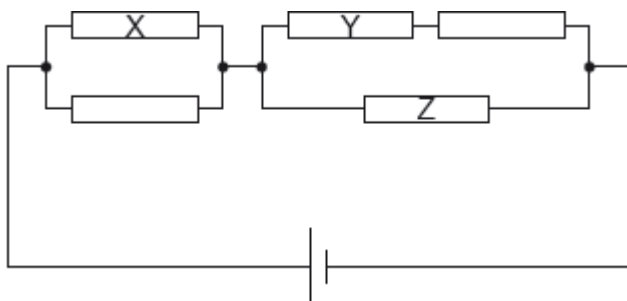


What is the resistance of R?

- A. $0.8\ \Omega$
- B. $2.0\ \Omega$
- C. $4.0\ \Omega$
- D. $8.0\ \Omega$

18M.1.SL.TZ1.20

Five resistors of equal resistance are connected to a cell as shown.



What is correct about the power dissipated in the resistors?

- A. The power dissipated is greatest in resistor X.
- B. The power dissipated is greatest in resistor Y.
- C. The power dissipated is greatest in resistor Z.

- D. The power dissipated is the same in all resistors.

17N.1.SL.TZ0.19

With reference to internal energy conversion and ability to be recharged, what are the characteristics of a primary cell?

	Internal energy conversion	Ability to be recharged
A.	chemical to electrical	rechargeable
B.	chemical to electrical	not rechargeable
C.	electrical to chemical	rechargeable
D.	electrical to chemical	not rechargeable

16N.1.HL.TZ0.17

A 12V battery has an internal resistance of 2.0Ω . A load of variable resistance is connected across the battery and adjusted to have resistance equal to that of the internal resistance of the battery. Which statement is correct for this circuit?

- A. The current in the battery is 6A.
B. The potential difference across the load is 12V.
C. The power dissipated in the battery is 18W.
D. The resistance in the circuit is 1.0Ω .

20N.1.HL.TZ0.18

An electrical power supply has an internal resistance. It supplies a direct current I to an external circuit for a time t . What is the electromotive force (emf) of the power supply?

- A. $\frac{\text{total energy transferred to the whole circuit}}{I \times t}$
B. $\frac{\text{total power transferred to the whole circuit}}{I \times t}$
C. $\frac{\text{total energy transferred to the external circuit}}{I \times t}$
D. $\frac{\text{total power transferred to the external circuit}}{I \times t}$

17M.1.SL.TZ1.21

An electron travelling at speed v perpendicular to a magnetic field of strength B experiences a force F .

What is the force acting on an alpha particle travelling at $2v$ parallel to a magnetic field of strength $2B$?

- A. 0
- B. $2F$
- C. $4F$
- D. $8F$

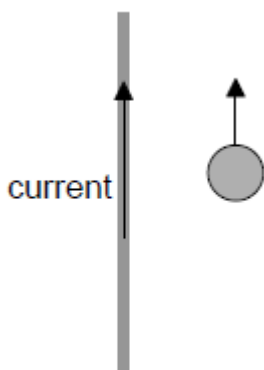
16N.1.SL.TZ0.21

A wire carrying a current I is at right angles to a uniform magnetic field of strength B . A magnetic force F is exerted on the wire. Which force acts when the same wire is placed at right angles to a uniform magnetic field of strength $2B$ when the current is $\frac{I}{4}$?

- A. $\frac{F}{4}$
- B. $\frac{F}{2}$
- C. F
- D. $2F$

17M.1.SL.TZ2.21

A positively-charged particle moves parallel to a wire that carries a current upwards.



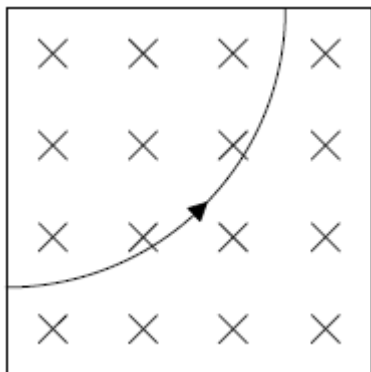
What is the direction of the magnetic force on the particle?

- A. To the left
- B. To the right
- C. Into the page

D. Out of the page

17M.1.HL.TZ2.17

The diagram shows the path of a particle in a region of uniform magnetic field. The field is directed into the plane of the page.

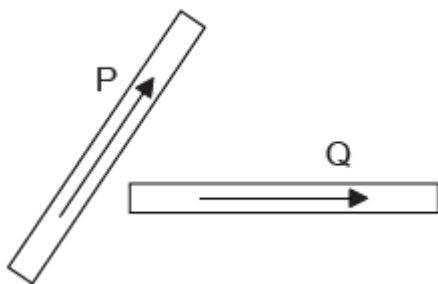


This particle could be

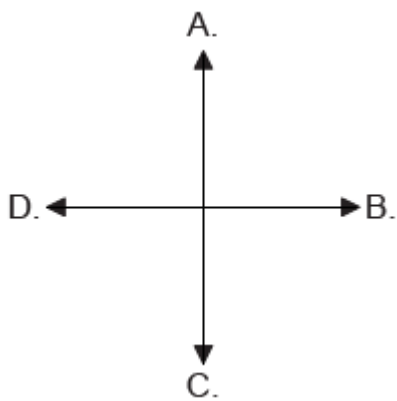
- A. an alpha particle.
- B. a beta particle.
- C. a photon.
- D. a neutron.

17N.1.SL.TZ0.20

The diagram shows two current-carrying wires, P and Q, that both lie in the plane of the paper. The arrows show the conventional current direction in the wires.



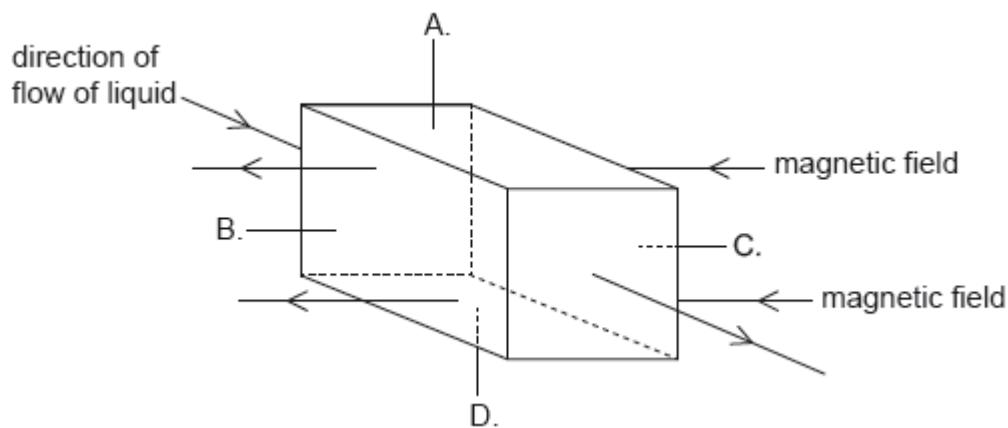
The electromagnetic force on Q is in the same plane as that of the wires. What is the direction of the electromagnetic force acting on Q?



18M.1.SL.TZ1.19

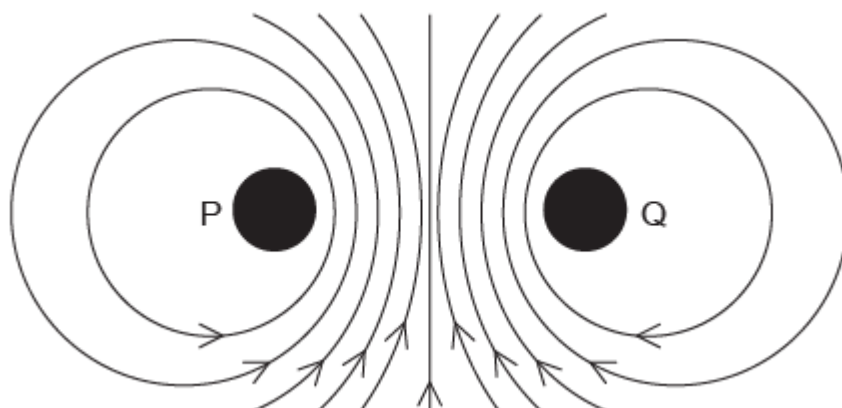
A liquid that contains negative charge carriers is flowing through a square pipe with sides A, B, C and D. A magnetic field acts in the direction shown across the pipe.

On which side of the pipe does negative charge accumulate?



17N.1.HL.TZ0.18

The diagram shows the magnetic field surrounding two current-carrying metal wires P and Q. The wires are parallel to each other and at right angles to the plane of the page.

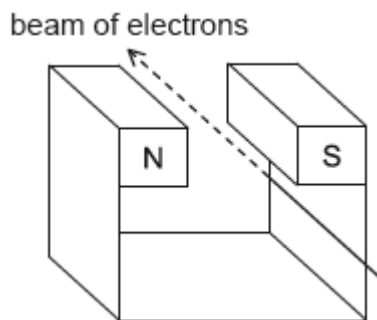


What is the direction of the electron flow in P and the direction of the electron flow in Q?

	Direction of electron flow in P	Direction of electron flow in Q
A.	into page	into page
B.	into page	out of page
C.	out of page	into page
D.	out of page	out of page

18M.1.SL.TZ2.21

A beam of electrons moves between the poles of a magnet.

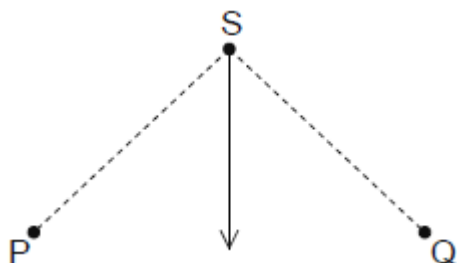


What is the direction in which the electrons will be deflected?

- A. Downwards
- B. Towards the N pole of the magnet
- C. Towards the S pole of the magnet
- D. Upwards

18N.1.HL.TZ0.18

Two parallel wires P and Q are perpendicular to the page and carry equal currents. Point S is the same distance from both wires. The arrow shows the magnetic field at S due to P and Q.

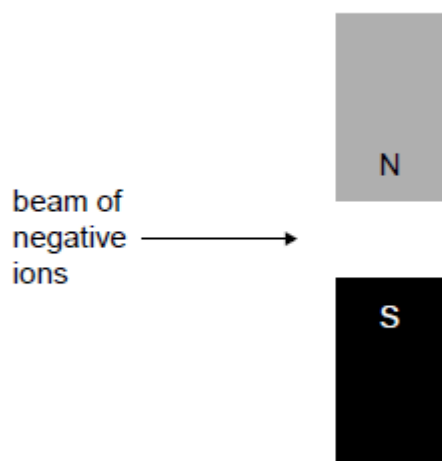


What are the correct directions for the current at P and the current at Q?

	Current direction at P	Current direction at Q
A.	into page	out of page
B.	out of page	out of page
C.	into page	into page
D.	out of page	into page

19M.1.SL.TZ1.23

A beam of negative ions flows in the plane of the page through the magnetic field due to two bar magnets.



What is the direction in which the negative ions will be deflected?

- A. Out of the page $\odot$
- B. Into the page $\otimes$
- C. Up the page $\uparrow$
- D. Down the page $\downarrow$

21M.1.HL.TZ1.18

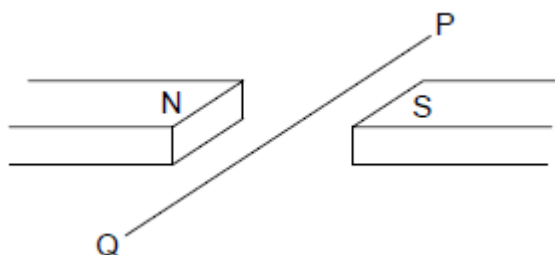
An electron enters the space inside a current-carrying solenoid. The velocity of the electron is parallel to the solenoid's axis. The electron is

- A. slowed down.

- B. speeded up.
- C. undeflected.
- D. deflected outwards.

19M.1.SL.TZ2.21

A horizontal wire PQ lies perpendicular to a uniform horizontal magnetic field.



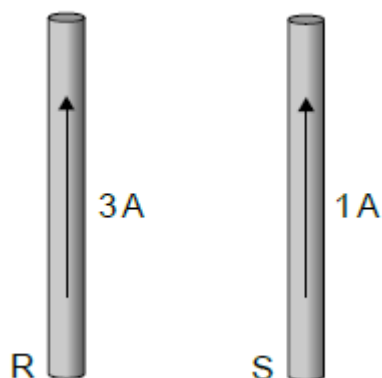
A length of 0.25 m of the wire is subject to a magnetic field strength of 40 mT. A downward magnetic force of 60 mN acts on the wire.

What is the magnitude and direction of the current in the wire?

	Current magnitude / A	Current direction
A.	6.0	P to Q
B.	6.0	Q to P
C.	0.17	Q to P
D.	0.17	P to Q

19M.1.HL.TZ1.18

Two currents of 3 A and 1 A are established in the same direction through two parallel straight wires R and S.



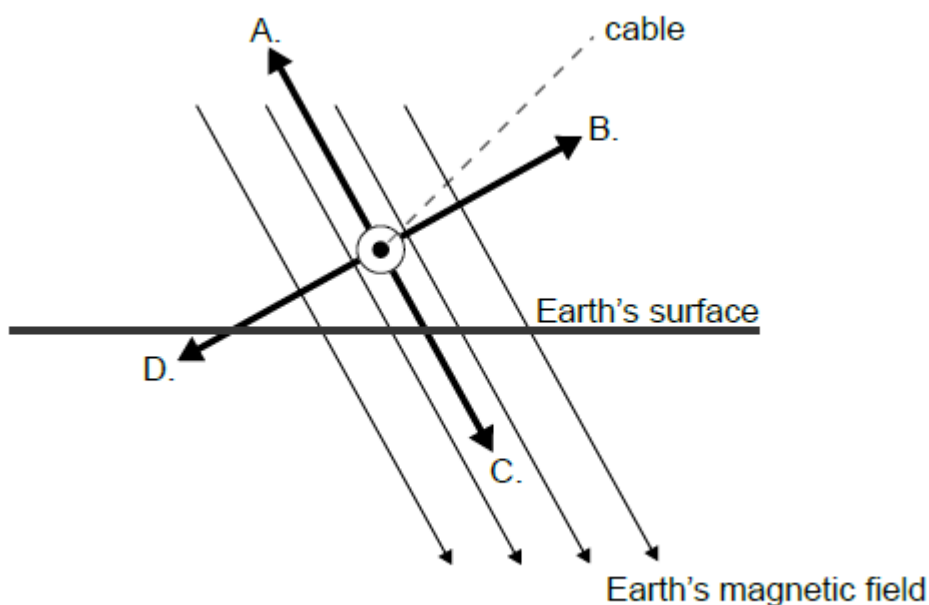
What is correct about the magnetic forces acting on each wire?

- A. Both wires exert equal magnitude attractive forces on each other.
- B. Both wires exert equal magnitude repulsive forces on each other.
- C. Wire R exerts a larger magnitude attractive force on wire S.
- D. Wire R exerts a larger magnitude repulsive force on wire S.

19M.1.HL.TZ1.19

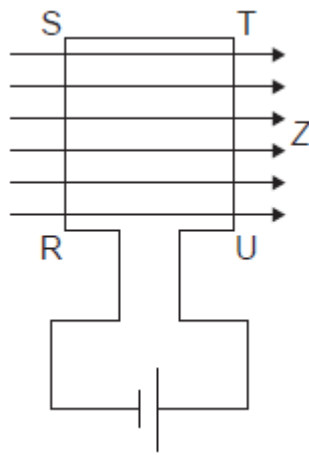
A horizontal electrical cable carries a steady current out of the page. The Earth's magnetic field exerts a force on the cable.

Which arrow shows the direction of the force on the cable due to the Earth's magnetic field?



22M.1.SL.TZ2.22

A rectangular coil of wire RSTU is connected to a battery and placed in a magnetic field Z directed to the right. Both the plane of the coil and the magnetic field direction are in the same plane.

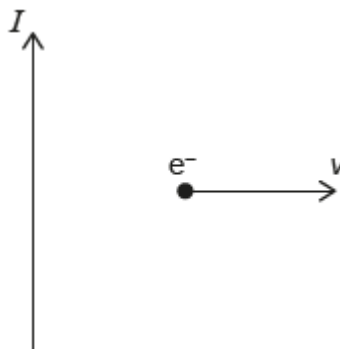


What is true about the magnetic force acting on the sides RS and ST?

	Force acting on RS	Force acting on ST
A.	into the page	into the page
B.	out of the page	no force acts
C.	into the page	no force acts
D.	out of the page	out of the page

21M.1.SL.TZ1.21

A long straight vertical conductor carries a current I upwards. An electron moves with horizontal speed v to the right.

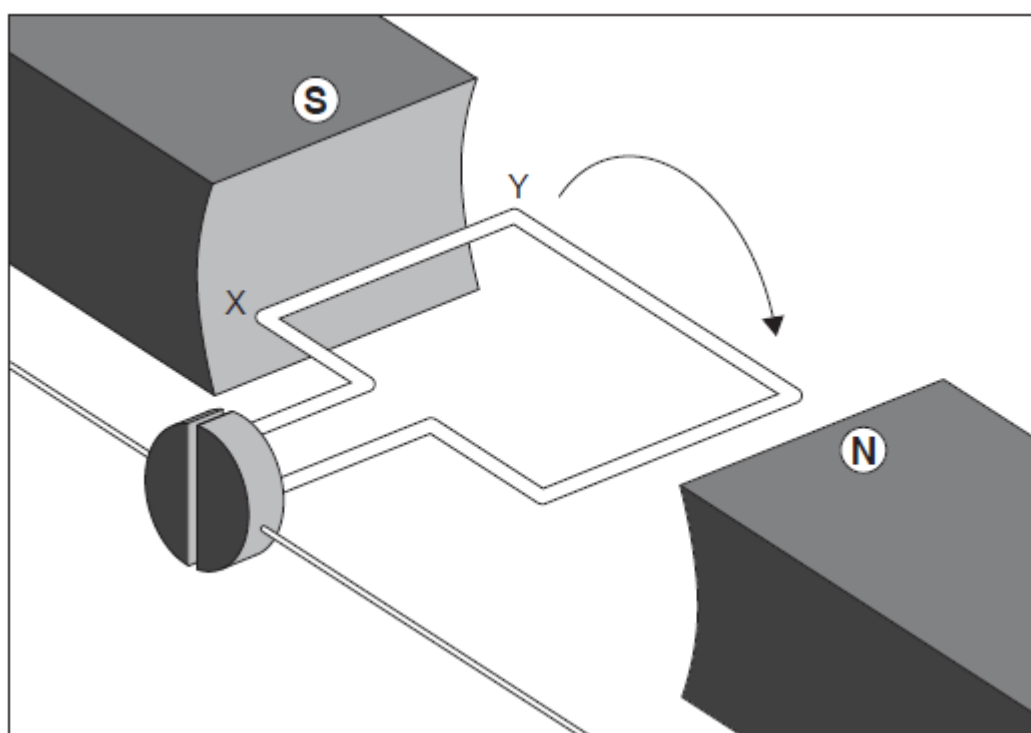


What is the direction of the magnetic force on the electron?

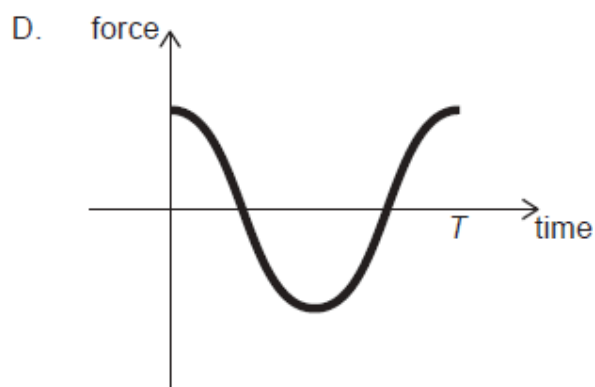
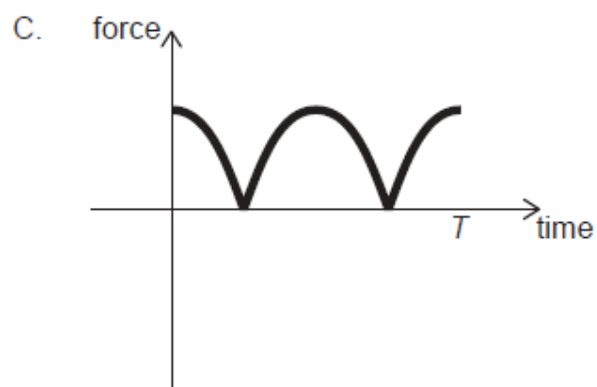
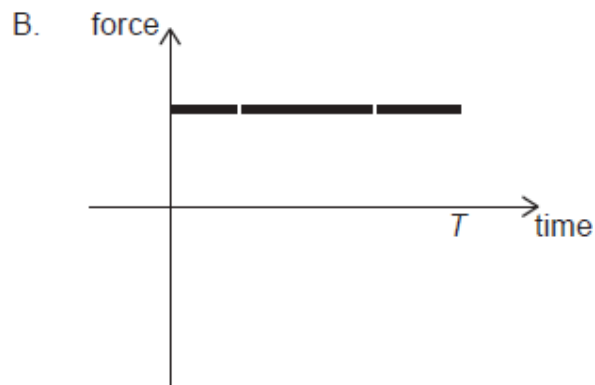
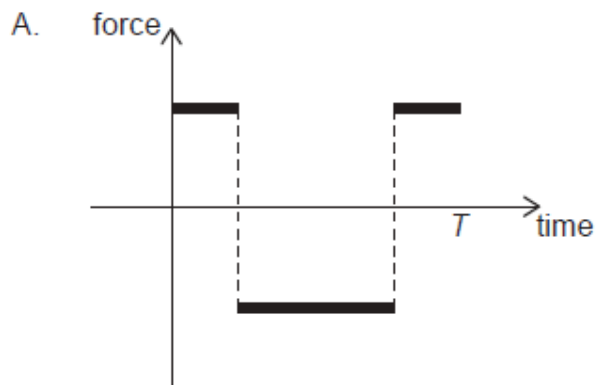
- A. Downwards
- B. Upwards
- C. Into the page
- D. Out of the page

22M.1.HL.TZ2.19

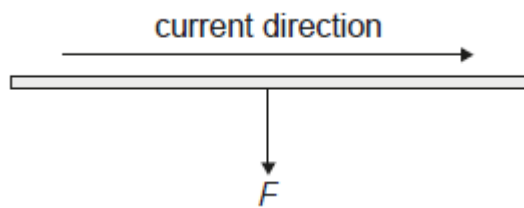
The coil of a direct current electric motor is turning with a period T . At $t = 0$ the coil is in the position shown in the diagram. Assume the magnetic field is uniform across the coil.



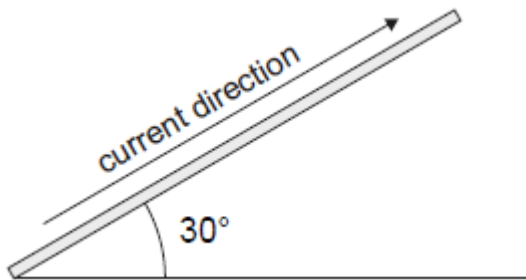
Which graph shows the variation with time of the force exerted on section XY of the coil during one complete turn?



A conductor is placed in a uniform magnetic field perpendicular to the plane of the paper. A force F acts on the conductor when there is a current in the conductor as shown.



The conductor is rotated 30° about the axis of the magnetic field.



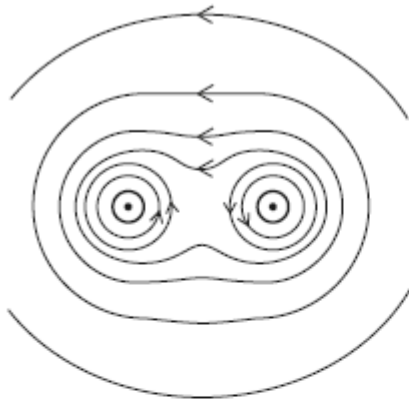
What is the direction of the magnetic field and what is the magnitude of the force on the conductor after the rotation?

	Direction of magnetic field	Magnitude of force
A.	into the plane of the paper	F
B.	into the plane of the paper	$\frac{F}{2}$
C.	out of the plane of the paper	F
D.	out of the plane of the paper	$\frac{F}{2}$

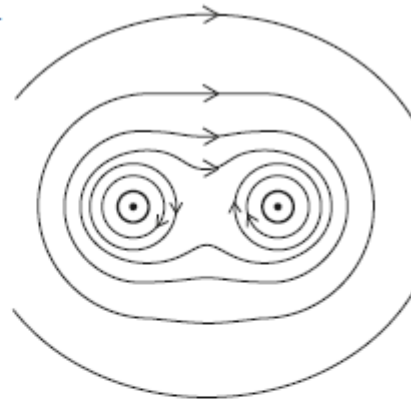
21N.1.HL.TZ0.18

Two parallel wires carry equal currents in the same direction out of the paper. Which diagram shows the magnetic field surrounding the wires?

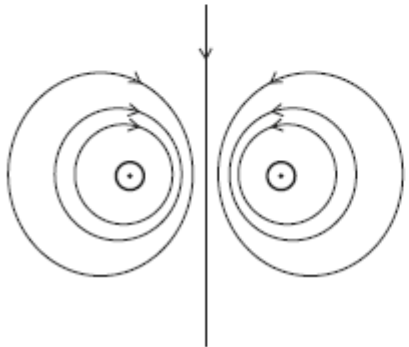
A.



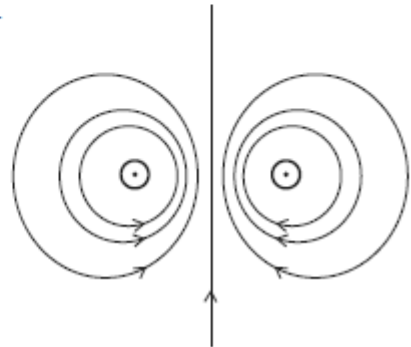
B.



C.



D.



21M.1.SL.TZ2.22

Magnetic field lines are an example of

- A. a discovery that helps us understand magnetism.
- B. a model to aid in visualization.
- C. a pattern in data from experiments.
- D. a theory to explain concepts in magnetism.

19N.1.SL.TZ0.20

When a wire with an electric current I is placed in a magnetic field of strength B it experiences a magnetic force F . What is the direction of F ?

- A. In a direction determined by I only
- B. In a direction determined by B only
- C. In the plane containing I and B
- D. At 90° to the plane containing I and B

Markschemes

19M.1.SL.TZ2.18

B

19M.1.SL.TZ2.20

D

19M.1.HL.TZ2.28

D

19M.1.HL.TZ2.29

C

19M.1.HL.TZ2.30

D

19M.1.HL.TZ2.31

D

19M.1.SL.TZ2.22

D

19M.1.SL.TZ2.24

A

19M.1.SL.TZ2.30

D

19N.1.HL.TZ0.34

B

19N.1.HL.TZ0.40

A

20N.1.SL.TZ0.21

B

20N.1.SL.TZ0.22

D

20N.1.SL.TZ0.23

C

20N.1.SL.TZ0.18

B

20N.1.SL.TZ0.24

C

20N.1.SL.TZ0.25

B

20N.1.SL.TZ0.19

C

20N.1.SL.TZ0.26

C

20N.1.SL.TZ0.20

A

20N.1.SL.TZ0.27

D

20N.1.SL.TZ0.28

A

20N.1.HL.TZ0.15

D

20N.1.HL.TZ0.16

D

20N.1.HL.TZ0.17

B

20N.1.HL.TZ0.24

A

20N.1.HL.TZ0.34

B

20N.1.HL.TZ0.30

D

20N.1.HL.TZ0.35

C

20N.1.HL.TZ0.31

C

20N.1.HL.TZ0.32

B

20N.1.HL.TZ0.33

C

19N.1.SL.TZ0.22

C

19N.1.SL.TZ0.21

A

22M.1.SL.TZ1.23

B

21M.1.SL.TZ2.24

D

21M.1.SL.TZ2.23

A

21M.1.SL.TZ2.19

C

21N.1.SL.TZ0.22

D

22M.1.SL.TZ2.23

D

19M.1.SL.TZ2.14

B

21M.1.SL.TZ1.22

B

19M.1.SL.TZ1.25

D

19M.1.SL.TZ1.24

A

18N.1.HL.TZ0.5

A

18N.1.SL.TZ0.22

B

18N.1.SL.TZ0.7

A

18M.1.HL.TZ2.17

D

18M.1.SL.TZ2.23

D

17N.1.SL.TZ0.22

B

18M.1.SL.TZ1.22

A

17N.1.SL.TZ0.21

B

17M.1.HL.TZ2.18

D

17M.1.SL.TZ2.22

A

16N.1.SL.TZ0.22

A

17M.1.SL.TZ1.23

A

17M.1.SL.TZ1.22

B

19M.1.SL.TZ2.28

B

17M.1.SL.TZ2.28

A

18M.1.HL.TZ2.23

A

18M.1.HL.TZ1.23

B

18N.1.SL.TZ0.28

D

19N.1.SL.TZ0.29

A

19N.1.HL.TZ0.6

A

17M.1.SL.TZ1.25

A

17M.1.SL.TZ1.26

A

17M.1.SL.TZ1.27

D

21M.1.HL.TZ2.22

A

16N.1.SL.TZ0.25

B

16N.1.SL.TZ0.26

C

17M.1.SL.TZ2.26

C

17M.1.HL.TZ2.21

C

18M.1.SL.TZ1.25

C

17N.1.SL.TZ0.24

B

18M.1.SL.TZ2.26

C

18N.1.SL.TZ0.25

A

19M.1.SL.TZ1.26

A

19M.1.SL.TZ1.27

B

21M.1.SL.TZ1.25

B

21M.1.HL.TZ1.22

A

19M.1.SL.TZ2.25

D

22M.1.HL.TZ2.24

D

21N.1.SL.TZ0.25

B

22M.1.HL.TZ2.38

A

22M.1.HL.TZ1.40

D

21M.1.HL.TZ1.40

D

21M.1.HL.TZ1.39

A

19M.1.HL.TZ1.39

C

18N.1.HL.TZ0.40

A

18N.1.HL.TZ0.39

D

18M.1.HL.TZ2.40

C

17M.1.HL.TZ2.40

A

17M.1.HL.TZ1.40

D

21M.1.HL.TZ2.34

B

16N.1.HL.TZ0.33

D

17M.1.HL.TZ1.32

D

16N.1.HL.TZ0.34

A

17N.1.HL.TZ0.36

B

18M.1.HL.TZ1.34

D

18M.1.HL.TZ2.35

A

18N.1.HL.TZ0.34

C

21M.1.HL.TZ1.35

A

22M.1.HL.TZ1.35

C

21N.1.HL.TZ0.33

C

19N.1.HL.TZ0.33

C

22M.1.HL.TZ1.34

A

22M.1.HL.TZ2.35

D

21M.1.HL.TZ1.34

B

21M.1.HL.TZ1.33

D

19M.1.HL.TZ1.35

B

18N.1.HL.TZ0.33

D

18M.1.HL.TZ2.34

A

18M.1.HL.TZ2.33

C

18M.1.HL.TZ1.33

C

17N.1.HL.TZ0.34

D

17M.1.HL.TZ2.34

A

17M.1.HL.TZ2.33

B

17M.1.HL.TZ1.36

D

17M.1.HL.TZ1.33

B

21M.1.HL.TZ2.35

A

21M.1.HL.TZ2.32

B

17M.1.HL.TZ1.30

C

16N.1.HL.TZ0.30

B

16N.1.HL.TZ0.32

A

17M.1.HL.TZ2.32

C

17M.1.HL.TZ2.31

B

17N.1.HL.TZ0.32

C

18M.1.HL.TZ2.31

C

18M.1.HL.TZ1.32

A

18M.1.HL.TZ2.32

C

18N.1.HL.TZ0.31

A

18N.1.HL.TZ0.32

D

19M.1.HL.TZ2.10

B

19M.1.HL.TZ2.11

A

19M.1.HL.TZ1.33

C

19M.1.HL.TZ1.34

C

21M.1.HL.TZ1.32

B

22M.1.HL.TZ2.32

A

22M.1.HL.TZ2.33

D

22M.1.HL.TZ1.33

B

21N.1.HL.TZ0.32

C

19N.1.HL.TZ0.30

D

19N.1.HL.TZ0.32

B

21N.1.HL.TZ0.30

D

22M.1.HL.TZ1.32

A

21M.1.HL.TZ1.30

D

19M.1.HL.TZ1.32

D

18M.1.HL.TZ1.31

D

18M.1.HL.TZ2.30

C

18M.1.HL.TZ2.29

A

18M.1.HL.TZ1.30

D

18M.1.HL.TZ2.28

C

17N.1.HL.TZ0.31

C

17N.1.HL.TZ0.33

B

17M.1.HL.TZ2.30

B

17M.1.HL.TZ1.31

B

17M.1.HL.TZ1.29

B

16N.1.HL.TZ0.31

B

21M.1.HL.TZ2.31

A

21M.1.HL.TZ2.30

A

17M.1.SL.TZ1.19

C

17M.1.SL.TZ1.18

C

17M.1.SL.TZ1.20

D

17M.1.HL.TZ1.17

D

22M.1.HL.TZ2.16

D

16N.1.SL.TZ0.18

C

17M.1.SL.TZ2.18

C

17M.1.SL.TZ2.19

B

17N.1.HL.TZ0.15

A

18M.1.HL.TZ1.15

C

18M.1.SL.TZ2.20

C

18N.1.SL.TZ0.18

A

18N.1.HL.TZ0.30

D

19M.1.SL.TZ1.22

D

21M.1.SL.TZ1.18

A

19M.1.SL.TZ2.2

C

19M.1.HL.TZ1.16

A

22M.1.SL.TZ2.19

A

21N.1.SL.TZ0.18

C

22M.1.SL.TZ1.19

B

22M.1.SL.TZ1.20

C

22M.1.SL.TZ1.22

B

19N.1.SL.TZ0.17

A

19N.1.SL.TZ0.18

D

19N.1.HL.TZ0.31

C

21N.1.HL.TZ0.17

C

21N.1.SL.TZ0.21

A

21M.1.SL.TZ1.20

A

18M.1.HL.TZ1.17

C

18M.1.SL.TZ2.22

D

18M.1.SL.TZ1.20

C

17N.1.SL.TZ0.19

B

16N.1.HL.TZ0.17

C

20N.1.HL.TZ0.18

A

17M.1.SL.TZ1.21

A

16N.1.SL.TZ0.21

B

17M.1.SL.TZ2.21

A

17M.1.HL.TZ2.17

A

17N.1.SL.TZ0.20

A

18M.1.SL.TZ1.19

A

17N.1.HL.TZ0.18

B

18M.1.SL.TZ2.21

D

18N.1.HL.TZ0.18

A

19M.1.SL.TZ1.23

A

21M.1.HL.TZ1.18

A

19M.1.SL.TZ2.21

B

19M.1.HL.TZ1.18

A

19M.1.HL.TZ1.19

B

22M.1.SL.TZ2.22

B

21M.1.SL.TZ1.21

A

22M.1.HL.TZ2.19

A

22M.1.HL.TZ1.22

C

21N.1.HL.TZ0.18

A

21M.1.SL.TZ2.22

B

19N.1.SL.TZ0.20

D